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JEE MAIN · CHEMISTRY · CHAPTER-WISE PYQS

Biomolecules

JEE Main Chemistry · Previous Year Questions with Answer Key

216	188	2002 – 2026	2	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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MCQ (Single Correct Answer)	191 questions
Numerical	23 questions
Out of syllabus	2 questions
Answer key	at the end

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MCQ (Single Correct Answer) 191 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

The incorrect statement from the following with respect to carbohydrates is :

(A) All monosaccharides are reducing sugars.

(B) The monosaccharide units obtained from hydrolysis of oligosaccharides are always the same.

(C) Starch and cellulose are typical examples of polysaccharides, which are very high molecular weight compounds of more than ten monosaccharide units.

(D) Open chain and cyclic structures co-exist at equilibrium that are responsible for certain properties as in the case of $D-(+)$-glucose.

Open solution

Q2. JEE Main 2026 (Online) 8th April Evening Shift

Which of the following amino acid will give violet coloured complex with neutral ferric chloride solution?

(A) Threonine (B) Serine (C) Tyrosine (D) Cysteine

Open solution

Q3. JEE Main 2026 (Online) 6th April Morning Shift

Match the List-I with List-II

List-I Deficiency Disease		List-II Vitamin	
A.	Scurvy	I.	Pyridoxine
B.	Convulsions	II.	Vitamin A
C.	Cheilosis	III.	Ascorbic Acid
D.	Xerophthalmia	IV.	Riboflavin

Choose the correct answer from the options given below :

(A) A-I, B-III, C-II, D-IV (B) A-I, B-III, C-IV, D-II

(C) A-III, B-I, C-IV, D-II (D) A-III, B-I, C-II, D-IV

Open solution

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Q4. JEE Main 2026 (Online) 6th April Morning Shift

Open solution

Match the List-I with List-II

List-I Amino acid		List-II Positive reaction/Test for functional group present in side chain of amino acid	
A.	Glutamine	I.	Hinsberg's test
B.	Lysine	II.	Neutral $FeCl_3$ test
C.	Tyrosine	III.	Ceric ammonium nitrate test
D.	Serine	IV.	Hoffman bromamide degradation

Choose the correct answer from the options given below :

(A) A-IV, B-II, C-I, D-III

(B) A-IV, B-I, C-II, D-III

(C) A-III, B-II, C-I, D-IV

(D) A-IV, B-I, C-III, D-II

Q5. JEE Main 2026 (Online) 5th April Evening Shift

Open solution

Identify the incorrect statement about tertiary structure of proteins.

(A) They can be fibrous or globular in structure.

(B) The main forces that stabilize the structure are hydrogen bonding, disulphide links, van der Waals and electrostatic forces of attraction.

(C) The structure remains intact when exposed to pH changes.

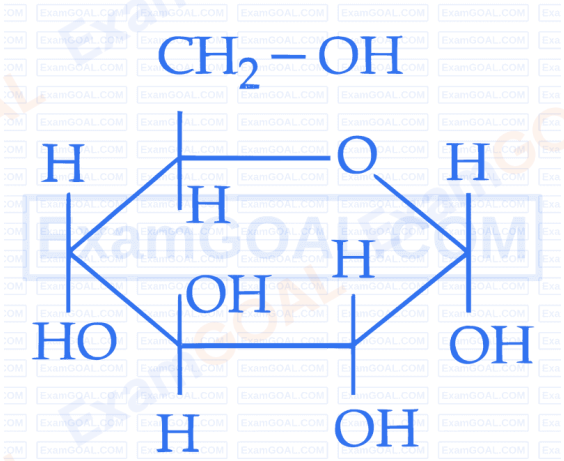
(D) A linear polypeptide chain will convert to a secondary structure and then further folding of the secondary structure will convert to tertiary structure.

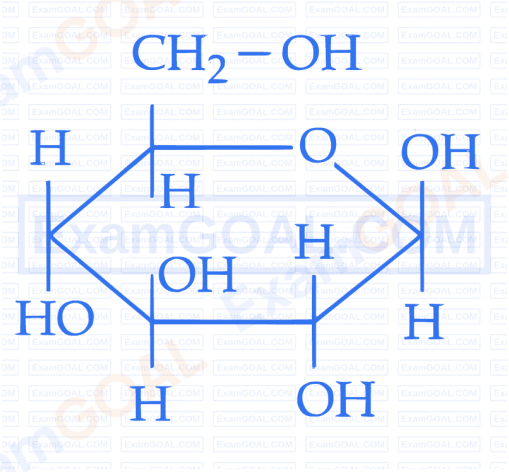
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Q6. JEE Main 2026 (Online) 5th April Evening Shift

Open solution

Given below are two statements :

Statement I :  and

 are two anomers of D-(+)-glucose.

Statement II : The open chain forms of D-glucose and D- fructose contain three similar chiral carbons at C_3, C_4 and C_5.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

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Q7. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

Identify the correct statements.

A. Glucose exists in two anomeric forms.

B. Anomers of glucose differ in configuration at $C-1$ in cyclic hemiacetal structure.

C. Melting point of α - anomer of glucose is greater than β - anomer.

D. Specific rotation of α - anomer is $+19^{\circ}$ while for β - anomer is $+112^{\circ}$

E. α and β - anomers of glucose are prepared by crystallization of saturated glucose solution at 303 K and 371 K respectively.

Choose the correct answer from the options given below :

(A) A and B Only **(B)** B and C Only **(C)** A, B and D Only **(D)** A, B and E Only

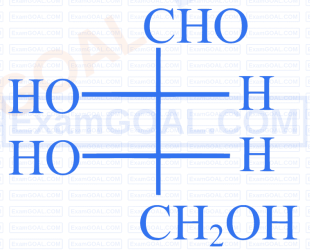
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Q8. JEE Main 2026 (Online) 4th April Evening Shift

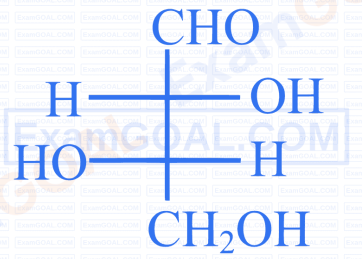
Open solution

A D-aldotetrose on oxidation with concentrated HNO_3 resulted in optically inactive dicarboxylic acid. The structure of the D-aldotetrose is :

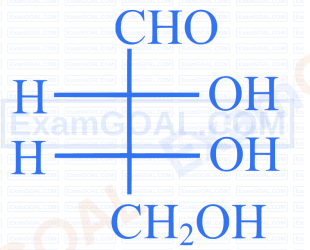
(A)



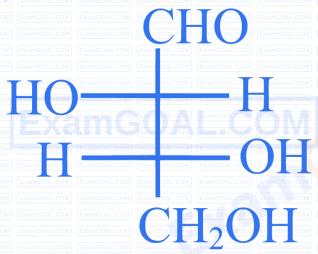
(B)



(C)



(D)



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Q9. JEE Main 2026 (Online) 4th April Morning Shift

Open solution

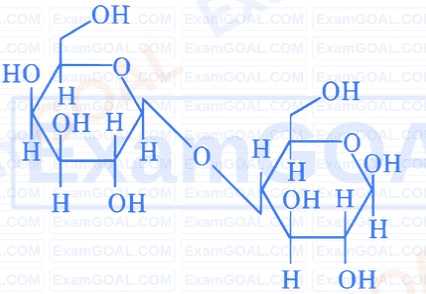
Given below are two statements :

Statement I : The structure of Maltose is given below:



Maltose is a non-reducing sugar.

Statement II : The structure of Lactose is given below :



Lactose is a reducing sugar.

In the light of the above statements, choose the **correct** answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q10. JEE Main 2026 (Online) 4th April Morning Shift

Open solution

Match the List-I with List-II

List-I **Name of amino acid**	**List-II** **One letter symbol/type**
A. Arginine	I. D/Non-essential
B. Aspartic acid	II. R/Essential
C. Lysine	III. E/Non-essential
D. Glutamic acid	IV. K/Essential

Choose the **correct** answer from the options given below :

(A) A-II, B-I, C-IV, D-III

(B) A-IV, B-III, C-II, D-I

(C) A-III, B-IV, C-I, D-II

(D) A-II, B-IV, C-I, D-III

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Q11. JEE Main 2026 (Online) 2nd April Evening Shift

Open solution

Identify the correct pair having amino acid (A) and the hormone (B) that is iodinated derivative of the amino acid (A).

(T and Y represent one letter code for amino acids)

(A) Amino acid (A) - T Hormone (B) - Insulin

(B) Amino acid (A) - T Hormone (B) - Thyroxine

(C) Amino acid (A) - Y Hormone (B) - Thyroxine

(D) Amino acid (A) - Y Hormone (B) - Insulin

Q12. JEE Main 2026 (Online) 2nd April Morning Shift

Open solution

Match List - I with List - II.

	List - I Vitamin		**List - II Name**
A	Vitamin B_1	I	Pyridoxine
B	Vitamin B_2	II	Ascorbic acid
C	Vitamin B_6	III	Thiamine
D	Vitamin C	IV	Riboflavin

Choose the **correct** answer from the options given below:

(A) A-II, B-I, C-III, D-IV

(B) A-IV, B-III, C-II, D-I

(C) A-III, B-IV, C-I, D-II

(D) A-I, B-III, C-II, D-IV

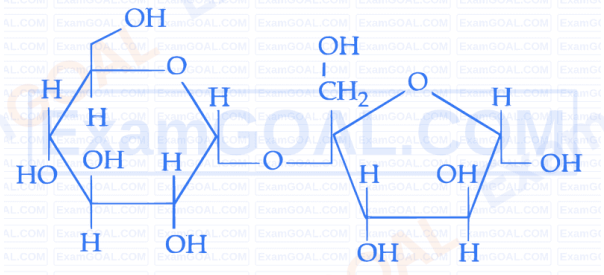
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Q13. JEE Main 2026 (Online) 28th January Evening Shift

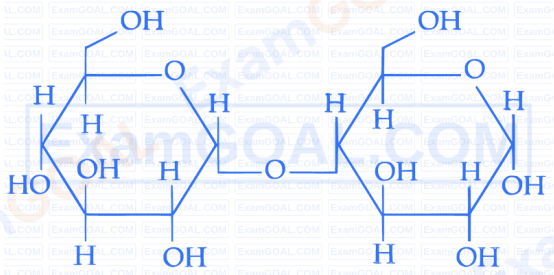
Open solution

Structures of four disaccharides are given below. Among the given disaccharides, the non-reducing sugar is :

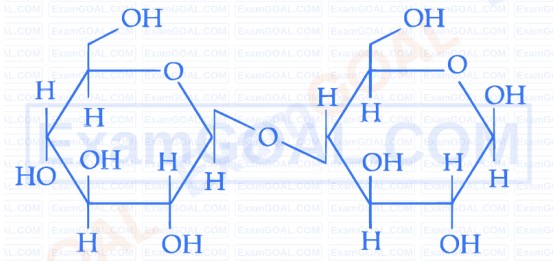
(A)



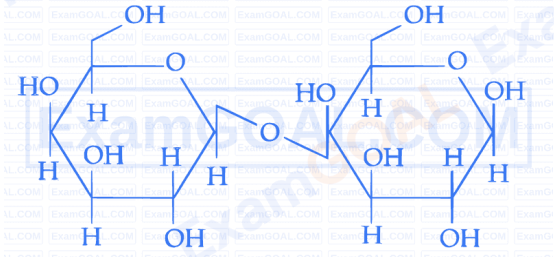
(B)



(C)



(D)

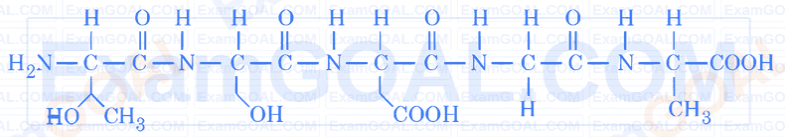


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Q14. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

In the given pentapeptide, find out an essential amino acid (Y) and the sequence present in the pentapeptide :



Choose the correct answer from the options given below :

(A) (Y) Serine (Sequence) Thr-Ser-Asp-Ala-Gly

(B) (Y) Serine (Sequence) Ser-Asp-Thr-Ala-Gly

(C) (Y) Threonine (Sequence) Ser-Thr-Asp-Gly-Ala

(D) (Y) Threonine (Sequence) Thr-Ser-Asp-Gly-Ala

Q15. JEE Main 2026 (Online) 24th January Evening Shift

Open solution

The number of possible tripeptides formed involving alanine (ala), glycine (gly) and valine (val), where no amino acid has been used more than once is:

(A) 8 **(B)** 3 **(C)** 4 **(D)** 6

Q16. JEE Main 2026 (Online) 23rd January Evening Shift

Open solution

Both human DNA and RNA are chiral molecules. The chirality in DNA and RNA arises due to the presence of

(A) L-sugar component **(B)** D-sugar component

(C) Chiral phosphate unit **(D)** Base unit

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Q17. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

Match List - I with List - II.

List - I Reaction of Glucose with		List - II Product formed	
A.	Hydroxylamine	I.	Gluconic acid
B.	Br_2 water	II.	Glucose pentaacetate
C.	Excess acetic anhydride	III.	Saccharic acid
D.	Concentrated HNO_3	IV.	Glucoxime

Choose the correct answer from the options given below :

(A) A-IV, B-III, C-II, D-I

(B) A-I, B-III, C-IV, D-II

(C) A-III, B-I, C-IV, D-II

(D) A-IV, B-I, C-II, D-III

Q18. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

Given below are two statements :

Statement I : Sucrose is dextrorotatory. However, sucrose upon hydrolysis gives a solution having mixture of products. This solution shows laevorotation.

Statement II : Hydrolysis of sucrose gives glucose and fructose. Since the laevorotation of glucose is more than the dextrorotation of fructose, the resulting solution becomes laevorotatory.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q19. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

The correct statements are :

A. Activation energy for enzyme catalysed hydrolysis of sucrose is lower than that of acid catalysed hydrolysis.

B. During denaturation, secondary and tertiary structures of a protein are destroyed but primary structure remains intact.

C. Nucleotides are joined together by glycosidic linkage between C_1 and C_4 carbons of the pentose sugar.

D. Quaternary structure of proteins represents overall folding of the polypeptide chain.

Choose the correct answer from the options given below :

(A) A, B and D Only

(B) A, C and D Only

(C) B and C Only

(D) A and B Only

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Q20. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

Identify the correct statements.

A. Arginine and Tryptophan are essential amino acids.

B. Histidine does not contain heterocyclic ring in its structure.

C. Proline is a six membered cyclic ring amino acid.

D. Glycine does not have chiral centre.

E. Cysteine has characteristic feature of side chain as $\mathrm{MeS}-\mathrm{CH_2}-\mathrm{CH_2}-$.

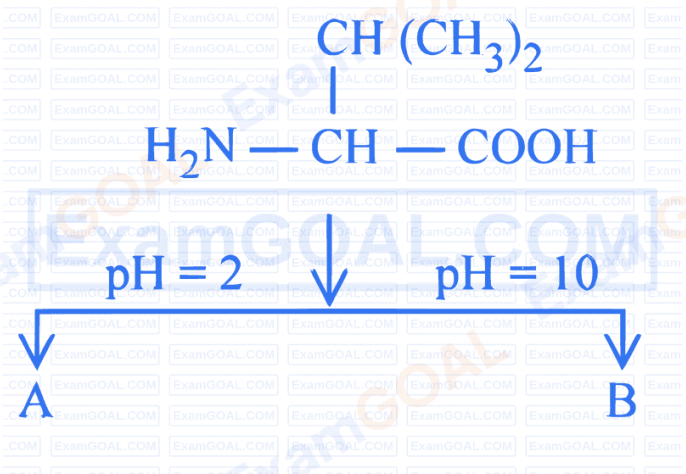
Choose the correct answer from the options given below :

(A) C and D Only **(B)** A and D Only **(C)** C and E Only **(D)** B and E Only

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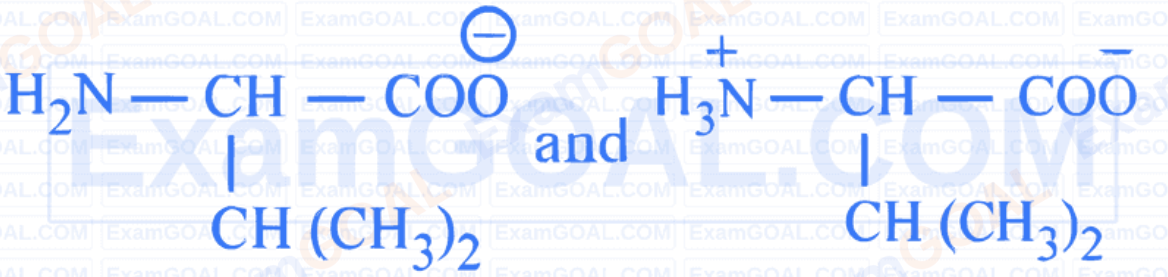
Q21. JEE Main 2025 (Online) 8th April Evening Shift

Open solution

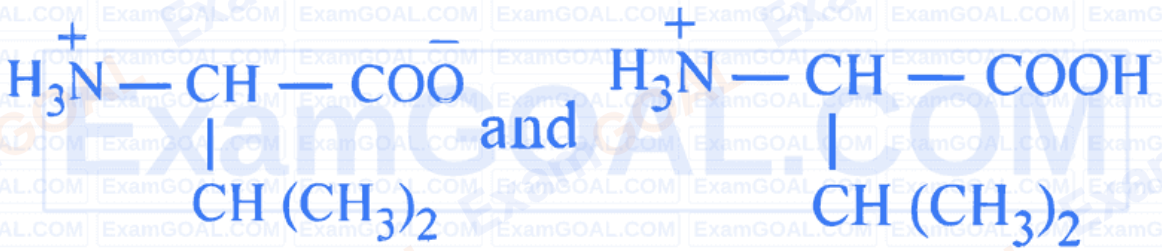


Choose the correct option for structures of A and B, respectively

(A)



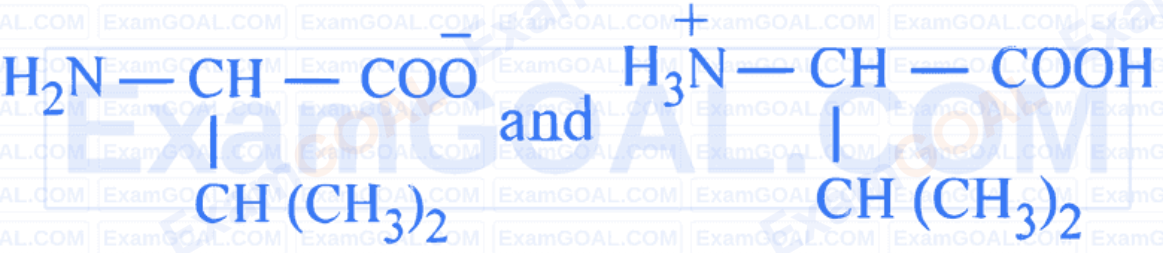
(B)



(C)



(D)



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Q22. JEE Main 2025 (Online) 7th April Evening Shift

Open solution

Given below are two statements :

Statement (I) : On hydrolysis, oligo peptides give rise to fewer number of α-amino acids while proteins give rise to a large number of β-amino acids.

Statement (II) : Natural proteins are denatured by acids which convert the water soluble form of fibrous proteins to their water insoluble form.

In the light of the above statements, choose the **most appropriate answer** from the options given below :

(A) **Statement I** is incorrect but **Statement II** is correct

(B) Both **Statement I** and **Statement II** are incorrect

(C) Both **Statement I** and **Statement II** are correct

(D) **Statement I** is correct but **Statement II** is incorrect

Q23. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

Given below are two statements:

Statement I: D-(+)-glucose + D-(+) fructose $\xrightarrow{-H_2O}$ Sucrose

sucrose $\xrightarrow{\text{hydrolysis}}$ $D-(+)$ glucose $+ D-(+)$ fructose

Statement II: Invert sugar is formed during sucrose hydrolysis

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are False

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

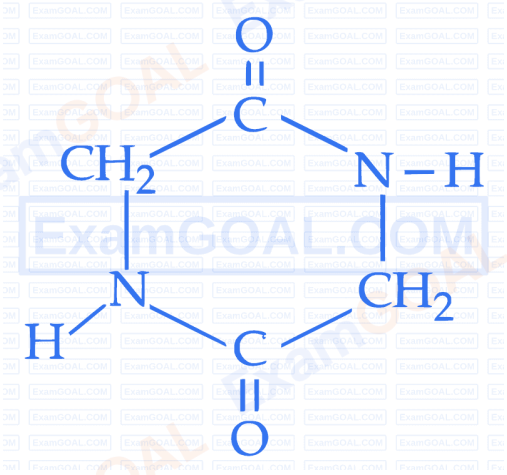
(D) Statement I is true but Statement II is false

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Q24. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

A dipeptide, " x " on complete hydrolysis gives " y " and " z ". " y " on treatment with aq. HNO_2 produces lactic acid. On the other hand " z " on heating gives the following cyclic molecule.



Based on the information given, the dipeptide X is :

(A) alanine-alanine (B) valine-glycine (C) valine-leucine (D) alanine-glycine

Q25. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

Identify the pair of reactants that upon reaction, with elimination of HCl will give rise to the dipeptide Gly-Ala.

(A) $NH_2\text{-}CH_2\text{-}COOH$ and $NH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-COOH$

(B) $NH_2\text{-}CH_2\text{-}COCl$ and $\overset{\oplus}{N}H_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-COCl$

(C) $NH_2\text{-}CH_2\text{-}COCl$ and $NH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-COOH$

(D) $NH_2\text{-}CH_2\text{-}COOH$ and $NH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-COCl$

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Q26. JEE Main 2025 (Online) 3rd April Evening Shift

Open solution

Fat soluble vitamins are:

A. Vitamin B_1

B. Vitamin C

C. Vitamin E

D. Vitamin B 12

E. Vitamin K

Choose the correct answer from the options given below:

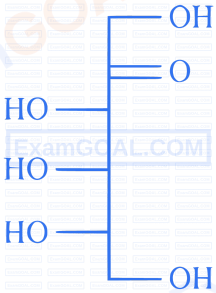
(A) B & C Only **(B)** C & D Only **(C)** C & E Only **(D)** A & B Only

Q27. JEE Main 2025 (Online) 3rd April Morning Shift

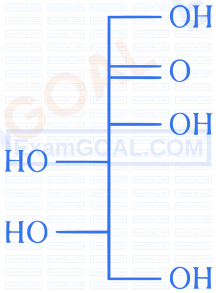
Open solution

Which of the following is the correct structure of L-Fructose?

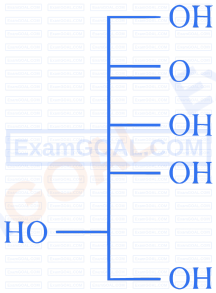
(A)



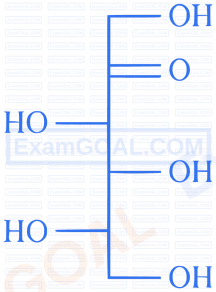
(B)



(C)



(D)



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Q28. JEE Main 2025 (Online) 2nd April Evening Shift

A tetrapeptide, " x " on complete hydrolysis produced glycine (Gly), alanine (Ala), valine (Val), leucine (Leu) in equimolar proportion each. The number of tetrapeptides (sequences) possible involving each of these amino acids is :

Open solution

(A) 32 **(B)** 24 **(C)** 8 **(D)** 16

Q29. JEE Main 2025 (Online) 2nd April Morning Shift

Identify the correct statement among the following :

Open solution

(A) All naturally occurring amino acids except glycine contain one chiral centre.

(B) Amino acid, cysteine can easily undergo dimerisation due to the presence of free SH group.

(C) Glutamic acid is the only amino acid that contains a $-$COOH group at the side chain.

(D) All naturally occurring amino acids are optically active.

Q30. JEE Main 2025 (Online) 29th January Evening Shift

Identify the essential amino acids from below:

Open solution

(A) Valine

(B) Proline

(C) Lysine

(D) Threonine

(E) Tyrosine

Choose the correct answer from the options given below:

(A) (A), (C) and (D) only **(B)** (C), (D) and (E) only

(C) (A), (C) and (E) only **(D)** (B), (C) and (E) only

Q31. JEE Main 2025 (Online) 29th January Morning Shift

Match List - I with List - II.

Open solution

List - I (Carbohydrate)	List - II (Linkage Source)
(A) Amylose	(I) β - $C_1 - C_4$, plant
(B) Cellulose	(II) α - $C_1 - C_4$, animal
(C) Glycogen	(III) α - $C_1 - C_4$, α - $C_1 - C_6$, plant
(D) Amylopectin	(IV) α - $C_1 - C_4$, plant

Choose the correct answer from the options given below :

(A) (A)-(II), (B)-(III), (C)-(I), (D)-(IV) **(B)** (A)-(III), (B)-(III), (C)-(I), (D)-(IV)

(C) (A)-(IV), (B)-(I), (C)-(III), (D)-(II) **(D)** (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

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Q32. JEE Main 2025 (Online) 28th January Evening Shift

Open solution

Match List-I with List-II.

List - I (Saccharides)	List - II (Glycosidic-linkages found)
(A) Sucrose	(I) α 1 – 4
(B) Maltose	(II) α 1 – 4 and α 1 – 6
(C) Lactose	(III) α 1 – β 2
(D) Amylopectin	(IV) β 1 – 4

Choose the **correct** answer from the options given below :

(A) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(B) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(C) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

(D) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

Q33. JEE Main 2025 (Online) 28th January Evening Shift

Open solution

Identify correct conversion during acidic hydrolysis from the following :

(A) starch gives galactose.

(B) cane sugar gives equal amount of glucose and fructose.

(C) milk sugar gives glucose and galactose.

(D) amylopectin gives glucose and fructose.

(E) amylose gives only glucose.

Choose the **correct** answer from the options given below :

(A) (A), (B) and (C) only

(B) (B), (C) and (D) only

(C) (B), (C) and (E) only

(D) (C), (D) and (E) only

Q34. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

Given below are two statements:

Statement I : D-glucose pentaacetate reacts with 2,4 -dinitrophenylhydrazine

Statement II : Starch, on heating with concentrated sulfuric acid at $100^{\circ}\mathrm{C}$ and $2-3\mathrm{atmosphere}$ pressure produces glucose.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

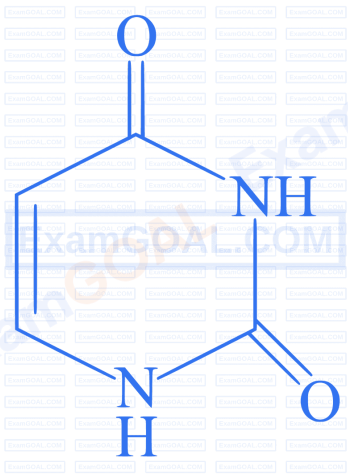
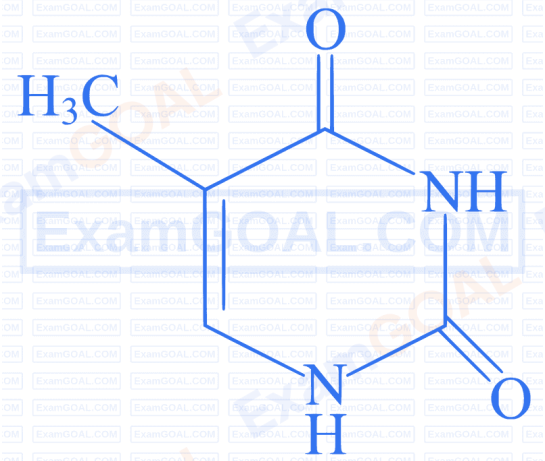
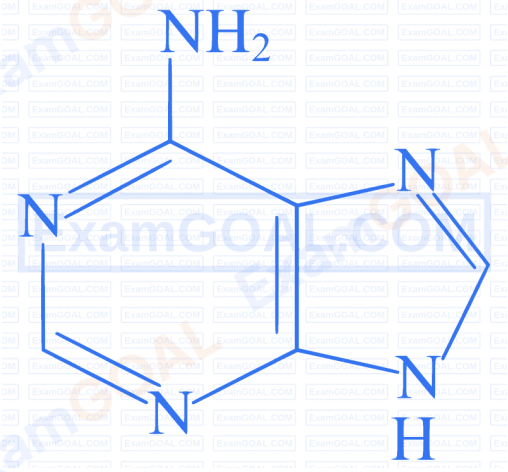
(D) Both Statement I and Statement II are false

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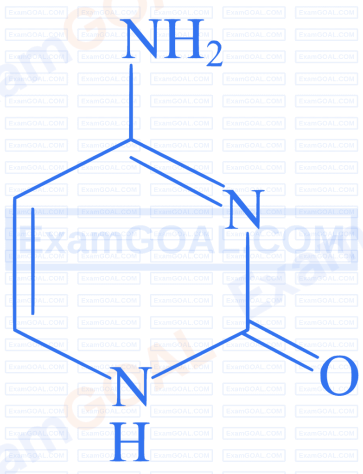
Q35. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

Match List - I with List - II.

	List - I		List - II
(A)	Adenine	(I)	
(B)	Cytosine	(II)	
(C)	Thymine	(III)	

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	List - I		List - II
(D)	Uracil	(IV)	

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(B) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

(C) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(D) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

Q36. JEE Main 2025 (Online) 24th January Morning Shift

The carbohydrate "Ribose" present in DNA, is

A. A pentose sugar

B. present in pyranose form

C. in "D" configuration

D. a reducing sugar, when free

E. in α-anomeric form

Choose the correct answer from the options given below:

(A) A, D and E Only **(B)** B, D and E Only **(C)** A, C and D Only **(D)** A, B and E Only

Open solution

Q37. JEE Main 2025 (Online) 23rd January Evening Shift

The α-Helix and β - Pleated sheet structures of protein are associated with its :

(A) tertiary structure

(B) quaternary structure

(C) primary structure

(D) secondary structure

Open solution

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Q38. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

Given below are two statements:

Statement I: Fructose does not contain an aldehydic group but still reduces Tollen's reagent

Statement II: In the presence of base, fructose undergoes rearrangement to give glucose.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

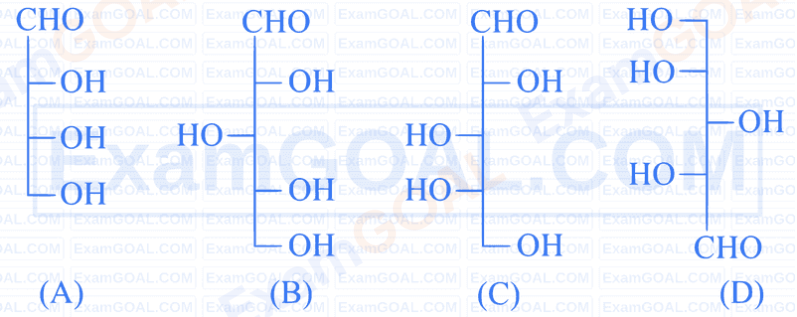
(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Q39. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

Identify the number of structure/s from the following which can be correlated to D-glyceraldehyde.



(A) three (B) two (C) one (D) four

Q40. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

Which of the following acids is a vitamin ?

(A) Saccharic acid (B) Ascorbic acid (C) Aspartic acid (D) Adipic acid

Q41. JEE Main 2024 (Online) 9th April Evening Shift

Open solution

The incorrect statement about Glucose is :

(A) Glucose remains in multiple isomeric form in its aqueous solution

(B) Glucose is soluble in water because of having aldehyde functional group

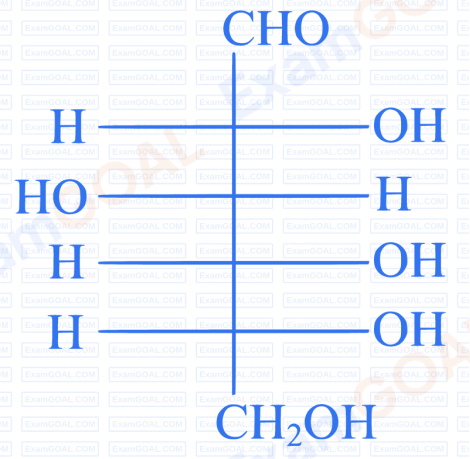
(C) Glucose is an aldohexose

(D) Glucose is one of the monomer unit in sucrose

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Q42. JEE Main 2024 (Online) 8th April Morning Shift

Open solution



The incorrect statement regarding the given structure is

(A) can be oxidized to a dicarboxylic acid with Br_2 water

(B) despite the presence of $-CHO$ does not give Schiff's test

(C) will coexist in equilibrium with 2 other cyclic structures

(D) has 4 asymmetric carbon atoms

Q43. JEE Main 2024 (Online) 6th April Evening Shift

Open solution

The incorrect statements regarding enzymes are :

(A) Enzymes are biocatalysts.

(B) Enzymes are non-specific and can catalyse different kinds of reactions.

(C) Most Enzymes are globular proteins.

(D) Enzyme - oxidase catalyses the hydrolysis of maltose into glucose.

Choose the correct answer from the option given below :

(A) (B) and (D) **(B)** (A), (B) and (C) **(C)** (B) and (C) **(D)** (B), (C) and (D)

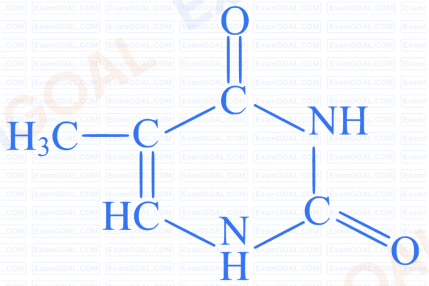
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Q44. JEE Main 2024 (Online) 6th April Morning Shift

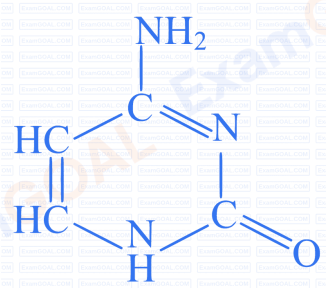
DNA molecule contains 4 bases whose structure are shown below. One of the structures is not correct, identify the incorrect base structure.

Open solution

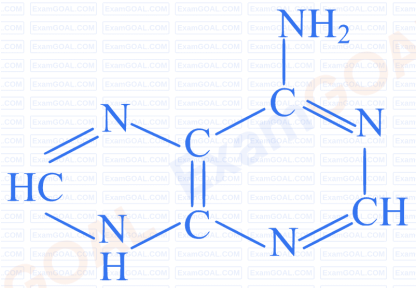
(A)



(B)



(C)



(D)



Q45. JEE Main 2024 (Online) 5th April Evening Shift

Coagulation of egg, on heating is because of :

Open solution

(A) Denaturation of protein occurs

(B) Biological property of protein remains unchanged

(C) Breaking of the peptide linkage in the primary structure of protein occurs

(D) The secondary structure of protein remains unchanged

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Q46. JEE Main 2024 (Online) 5th April Morning Shift

Which of the following gives a positive test with ninhydrin ?

(A) Polyvinyl chloride
(B) Cellulose
(C) Egg albumin
(D) Starch

Open solution

Q47. JEE Main 2024 (Online) 4th April Evening Shift

Match List I with List II

	LIST I		LIST II
A.	α - Glucose and α - Galactose	I.	Functional isomers
B.	α - Glucose and β - Glucose	II.	Homologous
C.	α - Glucose and α - Fructose	III.	Anomers
D.	α - Glucose and α - Ribose	IV.	Epimers

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-II, D-I
(B) A-IV, B-III, C-I, D-II
(C) A-III, B-IV, C-I, D-II
(D) A-IV, B-III, C-II, D-I

Open solution

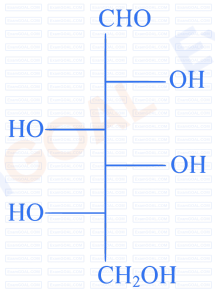
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Q48. JEE Main 2024 (Online) 4th April Morning Shift

Open solution

Which of the following is the correct structure of L-Glucose?

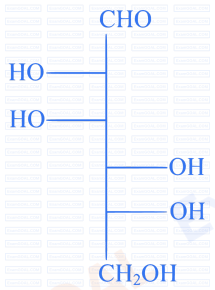
(A)



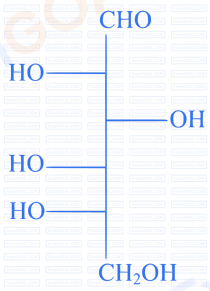
(B)



(C)



(D)



Q49. JEE Main 2024 (Online) 1st February Morning Shift

Open solution

If one strand of a DNA has the sequence ATGCTTCA, sequence of the bases in complementary strand is :

(A) CATTAGCT (B) TACGAAGT (C) ATGCGACT (D) GTACTTAC

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Q50. JEE Main 2024 (Online) 31st January Morning Shift

Open solution

Match List I with List II

	List - I		List - II
(A)	Glucose/$NaHCO_3/\Delta$	(I)	Gluconic acid
(B)	Glucose $/HNO_3$	(II)	No reaction
(C)	Glucose $/HI/\Delta$	(III)	n-hexane
(D)	Glucose/Bromine water	(IV)	Saccharic acid

Choose the correct answer from the options given below:

(A) A-II, B-IV, C-III, D-I **(B)** A-III, B-II, C-I, D-IV

(C) A-IV, B-I, C-III, D-II **(D)** A-I, B-IV, C-III, D-II

Q51. JEE Main 2024 (Online) 30th January Morning Shift

Open solution

Sugar which does not give reddish brown precipitate with Fehling's reagent, is :

(A) Glucose **(B)** Sucrose **(C)** Lactose **(D)** Maltose

Q52. JEE Main 2024 (Online) 29th January Evening Shift

Open solution

Match List I with List II

	List - I (Bio Polymer)		List - II (Monomer)
(A)	Starch	(I)	nucleotide
(B)	Cellulose	(II)	α-glucose
(C)	Nucleic acid	(III)	β-glucose
(D)	Protein	(IV)	α-amino acid

Choose the correct answer from the options given below:

(A) A-IV, B-II, C-I, D-III **(B)** A-I, B-III, C-IV, D-II

(C) A-II, B-I, C-III, D-IV **(D)** A-II, B-III, C-I, D-IV

Q53. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

Type of amino acids obtained by hydrolysis of proteins is :

(A) γ **(B)** β **(C)** α **(D)** δ

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Q54. JEE Main 2024 (Online) 27th January Evening Shift

Which structure of protein remains intact after coagulation of egg white on boiling?

(A) Quaternary (B) Primary (C) Secondary (D) Tertiary

Open solution

Q55. JEE Main 2024 (Online) 27th January Morning Shift

Two nucleotides are joined together by a linkage known as :

(A) Phosphodiester linkage (B) Glycosidic linkage

(C) Disulphide linkage (D) Peptide linkage

Open solution

Q56. JEE Main 2023 (Online) 15th April Morning Shift

Which is not true for arginine?

(A) It has a fairly high melting point.

(B) It is associated with more than one pK_a values.

(C) It has high solubility in benzene.

(D) It is a crystalline solid.

Open solution

Q57. JEE Main 2023 (Online) 13th April Evening Shift

The naturally occurring amino acid that contains only one basic functional group in its chemical structure is

(A) asparagine (B) arginine (C) histidine (D) lysine

Open solution

Q58. JEE Main 2023 (Online) 13th April Morning Shift

$$\text{D -(+)- Glyceraldehyde} \xrightarrow[\text{ii) } H_2O / H^+]{\text{i) HCN}} \xrightarrow{\text{ii) } HNO_3}$$

The products formed in the above reaction are

(A) Two optically active products

(B) One optically active and one meso product

(C) Two optically inactive products

(D) One optically inactive and one meso product.

Open solution

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Q59. JEE Main 2023 (Online) 11th April Morning Shift

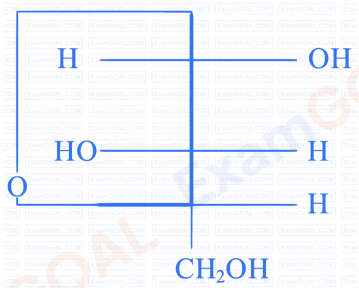
Open solution

L-isomer of tetrose X $(C_4H_8O_4)$ gives positive Schiff's test and has two chiral carbons. On acetylation, 'X' yields triacetate. 'X' also undergoes following reactions

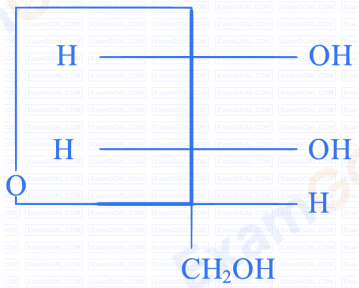


'X' is

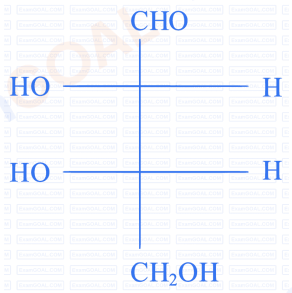
(A)



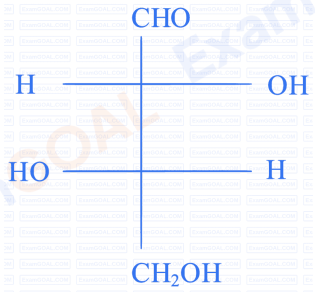
(B)



(C)



(D)



Q60. JEE Main 2023 (Online) 10th April Morning Shift

Open solution

The one that does not stabilize 2° and 3° structures of proteins is

(A) –O–O–linkage

(B) –S–S–linkage

(C) van der Waals forces

(D) H-bonding

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Q61. JEE Main 2023 (Online) 8th April Evening Shift

Open solution

Match List I with List II

	LIST I Natural amino acid		LIST II One letter code
A.	Glutamic acid	I.	Q
B.	Glutamine	II.	W
C.	Tyrosine	III.	E
D.	Tryptophan	IV.	Y

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-I, D-II

(B) A-IV, B-III, C-I, D-II

(C) A-II, B-I, C-IV, D-III

(D) A-III, B-I, C-IV, D-II

Q62. JEE Main 2023 (Online) 8th April Morning Shift

Open solution

Sulphur (S) containing amino acids from the following are:

(a) isoleucine (b) cysteine (c) lysine (d) methionine (e) glutamic acid

(A) b, d

(B) a, d

(C) b, c, e

(D) a, b, c

Q63. JEE Main 2023 (Online) 6th April Evening Shift

Open solution

Match List I with List II

	LIST I Natural Amino Acid		LIST II One Letter Code
A.	Arginine	I.	D
B.	Aspartic acid	II.	N
C.	Asparagine	III.	A
D.	Alanine	IV.	R

Choose the correct answer from the options given below:

(A) (A) - IV, (B) - I, (C) - III, (D) - II

(B) (A) - III, (B) - I, (C) - II, (D) - IV

(C) (A) - IV, (B) - I, (C) - II, (D) - III

(D) (A) - I, (B) - III, (C) - IV, (D) - II

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Q64. JEE Main 2023 (Online) 6th April Morning Shift

Open solution

Match List I with List II

	LIST I Enzymatic reaction		**LIST II Enzyme**
A.	Sucrose $\rightarrow$ Glucose and Fructose	I.	Zymase
B.	Glucose $\rightarrow$ ethyl alcohol and CO_2	II.	Pepsin
C.	Starch $\rightarrow$ Maltose	III.	Invertase
D.	Proteins $\rightarrow$ Amino acids	IV.	Diastase

Choose the correct answer from the options given below:

(A) A-III, B-I, C-II, D-IV

(B) A-III, B-I, C-IV, D-II

(C) A-I, B-IV, C-III, D-II

(D) A-I, B-II, C-IV, D-III

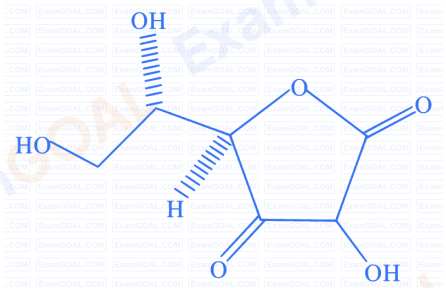
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Q65. JEE Main 2023 (Online) 1st February Evening Shift

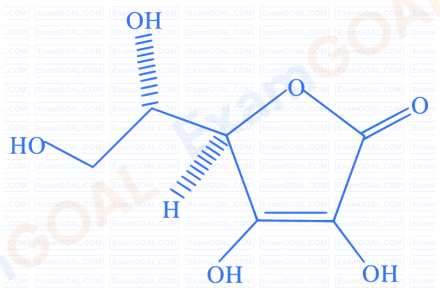
Open solution

All structures given below are of vitamin C. Most stable of them is :

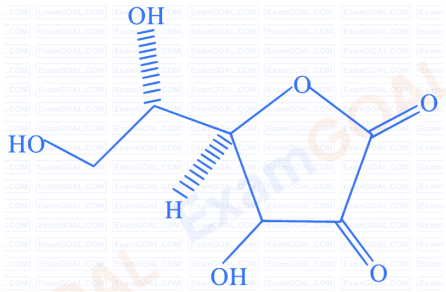
(A)



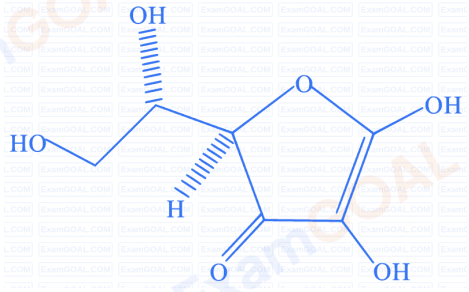
(B)



(C)



(D)



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Q66. JEE Main 2023 (Online) 1st February Evening Shift

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : α-halocarboxylic acid on reaction with dil. NH_3 gives good yield of α-amino carboxylic acid whereas the yield of amines is very low when prepared from alkyl halides.

Reason (R) : Amino acids exist in zwitter ion form in aqueous medium.

In the light of the above statements, choose the correct answer from the options given below :

(A) (A) is not correct but (R) is correct

(B) (A) is correct but (R) is not correct

(C) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

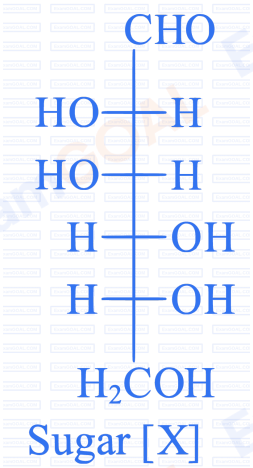
(D) Both (A) and (R) are correct and (R) is the correct explanation of (A)

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Q67. JEE Main 2023 (Online) 1st February Morning Shift

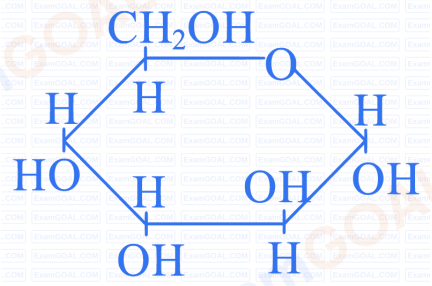
Open solution

The correct representation in six membered pyranose form for the following sugar [X] is

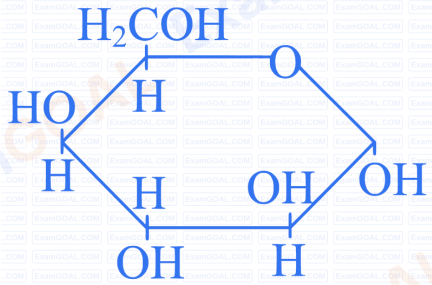


Sugar [X]

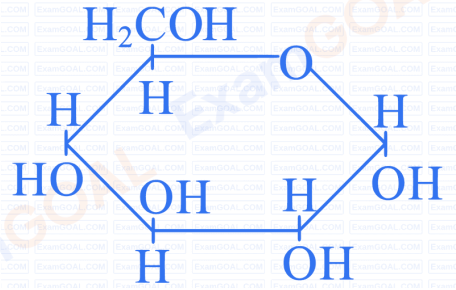
(A)



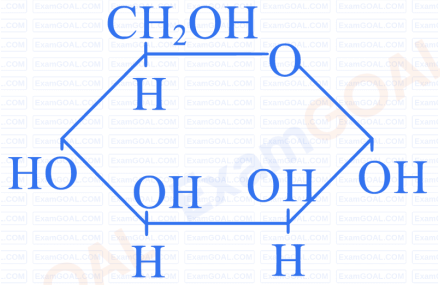
(B)



(C)



(D)



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Q68. JEE Main 2023 (Online) 1st February Morning Shift

Open solution

Match List I with List II

	List I		List II
	Test		Functional group / Class of Compound
A.	Molisch's Test	I.	Peptide
B.	Biuret Test	II.	Carbohydrate
C.	Carbylamine Test	III.	Primary amine
D.	Schiff's Test	IV.	Aldehyde

Choose the correct answer from the options given below :

(A) (A) - III, (B) - IV, (C) - II, (D) - I

(B) (A) - II, (B) - I, (C) - III, (D) - IV

(C) (A) - III, (B) - IV, (C) - I, (D) - II

(D) (A) - I, (B) - II, (C) - III, (D) - IV

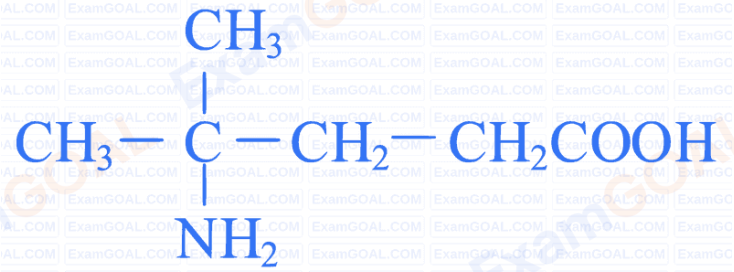
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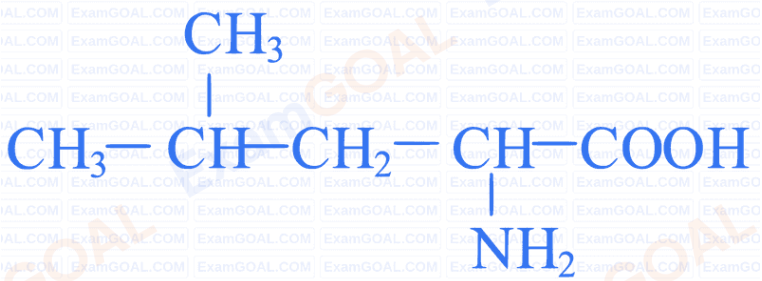
Q69. JEE Main 2023 (Online) 31st January Morning Shift

Open solution

A protein 'X' with molecular weight of $70,000\ \mathrm{u}$, on hydrolysis gives amino acids. One of these amino acids is

(A) 

(B) 

(C) 

(D) 

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Q70. JEE Main 2023 (Online) 30th January Morning Shift

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Ketoses give Seliwanoff's test faster than Aldoses.

Reason (R) : Ketoses undergo β-elimination followed by formation of furfural.

In the light of the above statements, choose the correct answer from the options given below :

(A) (A) is false but (R) is true

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

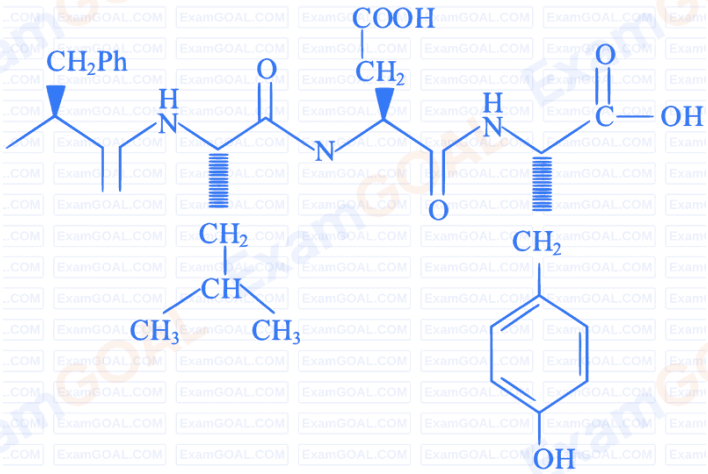
(C) (A) is true but (R) is false

(D) Both (A) and (R) are true and (R) is the correct explanation of (A)

Q71. JEE Main 2023 (Online) 29th January Evening Shift

Open solution

Following tetrapeptide can be represented as



(F, L, D, Y, I, Q, P are one letter codes for amino acids)

(A) YQLF **(B)** FIQY **(C)** PLDY **(D)** FLDY

Q72. JEE Main 2023 (Online) 29th January Morning Shift

Open solution

Number of cyclic tripeptides formed with 2 amino acids A and B is :

(A) 2 **(B)** 3 **(C)** 5 **(D)** 4

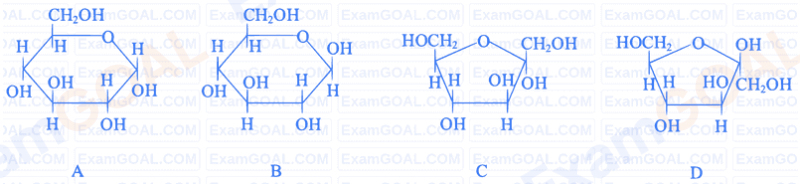
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Q73. JEE Main 2023 (Online) 25th January Morning Shift

Open solution

Match items of Row I with those of Row II.

Row I :



Row II :

(i) α-D-(−)-Fructofuranose

(ii) β-D-(−)-Fructofuranose

(iii) α-D-(−)-Glucopyranose

(iv) β-D-(−)-Glucopyranose

Correct match is

(A) A → iii, B → iv, C → i, D → ii

(B) A → iv, B → iii, C → i, D → ii

(C) A → i, B → ii, C → iii, D → iv

(D) A → iii, B → iv, C → ii, D → i

Q74. JEE Main 2022 (Online) 30th July Evening Shift

Open solution

Given below are two statements:

Statement I: Glycogen is highly branched polysaccharide and commonly known as animal starch due to its similarity with amylopectin.

Statement II: Amylose is water insoluble and constitutes about 15-20% of starch.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

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Q75. JEE Main 2022 (Online) 29th July Evening Shift

Open solution

Given below are two statements. One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Amylose is insoluble in water.

Reason R: Amylose is a long linear molecule with more than 200 glucose units.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are correct and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$.

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are correct but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$.

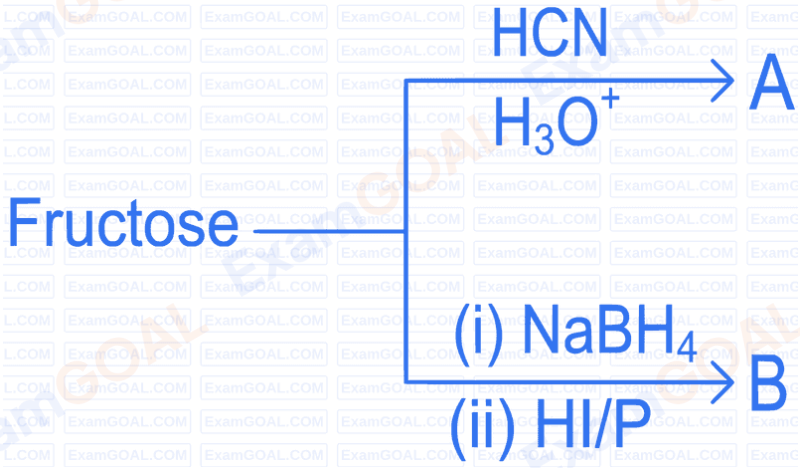
(C) $\mathbf{A}$ is correct but $\mathbf{R}$ is not correct

(D) $\mathbf{A}$ is not correct but $\mathbf{R}$ is correct.

Q76. JEE Main 2022 (Online) 28th July Evening Shift

Open solution

The formulas of A and B for the following reaction sequence



are

(A) $\mathrm{A} = \mathrm{C_7H_{14}O_8}, \quad \mathrm{B} = \mathrm{C_6H_{14}}$

(B) $\mathrm{A} = \mathrm{C_7H_{13}O_7}, \quad \mathrm{B} = \mathrm{C_7H_{14}O}$

(C) $\mathrm{A} = \mathrm{C_7H_{12}O_8}, \quad \mathrm{B} = \mathrm{C_6H_{14}}$

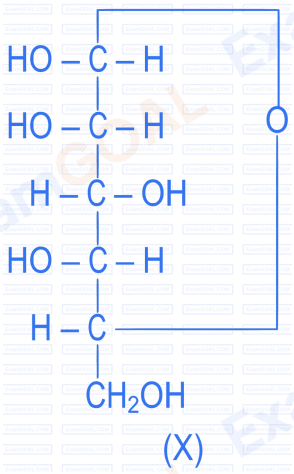
(D) $\mathrm{A} = \mathrm{C_7H_{14}O_8}, \quad \mathrm{B} = \mathrm{C_6H_{14}O_6}$

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Q77. JEE Main 2022 (Online) 28th July Morning Shift

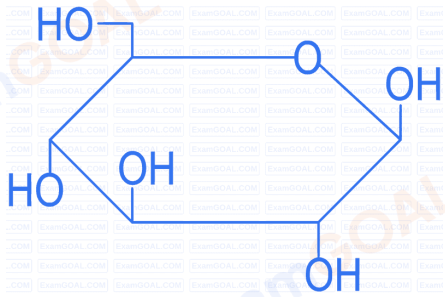
Open solution

For the below given cyclic hemiacetal (X), the correct pyranose structure is :

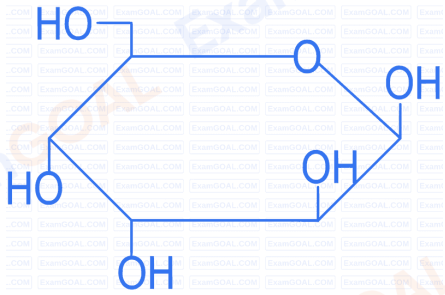


(X)

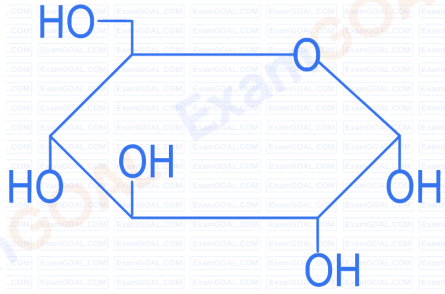
(A)



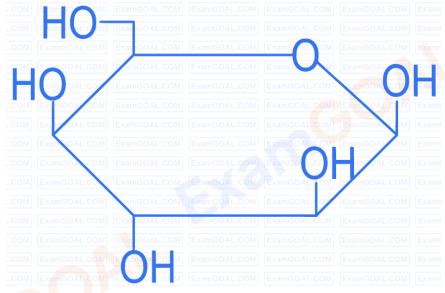
(B)



(C)



(D)



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Q78. JEE Main 2022 (Online) 27th July Evening Shift

Open solution

Match List - I with Match List - II.

	List - I		List - II
(A)	Glucose + HI	(I)	Gluconic acid
(B)	Glucose + Br_2 water	(II)	Glucose pentacetate
(C)	Glucose + acetic anhydride	(III)	Saccharic acid
(D)	Glucose + HNO_3	(IV)	Hexane

Choose the correct answer from the options given below:

(A) (A) - (IV), (B) - (I), (C) -(II), (D) - (III)

(B) (A) - (IV), (B) - (III), (C) -(II), (D) - (I)

(C) (A) - (III), (B) - (I), (C) -(IV), (D) - (II)

(D) (A) - (I), (B) - (III), (C) -(IV), (D) - (II)

Q79. JEE Main 2022 (Online) 27th July Morning Shift

Open solution

A sugar 'X' dehydrates very slowly under acidic condition to give furfural which on further reaction with resorcinol gives the coloured product after sometime. Sugar 'X' is

(A) Aldopentose **(B)** Aldotetrose **(C)** Oxalic acid **(D)** Ketotetrose

Q80. JEE Main 2022 (Online) 26th July Evening Shift

Open solution

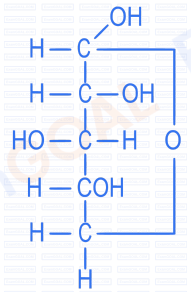
Animal starch is the other name of

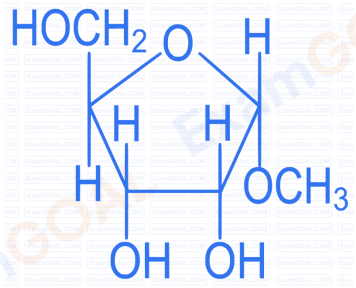
(A) amylose. **(B)** maltose. **(C)** glycogen. **(D)** amylopectin.

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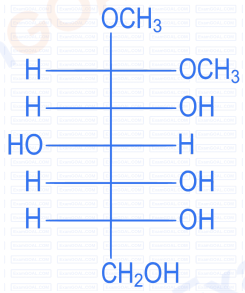
Q81. JEE Main 2022 (Online) 26th July Morning Shift

Which one of the following is a reducing sugar?

(A) 

(B) 

(C) 

(D) 

Open solution

Q82. JEE Main 2022 (Online) 25th July Evening Shift

Glycosidic linkage between C_1 of α-glucose and C_2 of β-fructose is found in

(A) maltose (B) sucrose (C) lactose (D) amylose

Open solution

Q83. JEE Main 2022 (Online) 25th July Morning Shift

During the denaturation of proteins, which of these structures will remain intact?

(A) Primary (B) Secondary (C) Tertiary (D) Quaternary

Open solution

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Q84. JEE Main 2022 (Online) 30th June Morning Shift

The sugar produced after complete hydrolysis of DNA is

(A) a pentose sugar.
(B) a hexose sugar.
(C) a polysaccharide.
(D) a disaccharide.

Open solution

Q85. JEE Main 2022 (Online) 29th June Evening Shift

The structure of protein that is unaffected by heating is

(A) secondary structure
(B) tertiary structure
(C) primary structure
(D) quaternary structure

Open solution

Q86. JEE Main 2022 (Online) 29th June Morning Shift

Sugar moiety in DNA and RNA molecules respectively are

(A) β-D-2-deoxyribose, β-D-deoxyribose.
(B) β-D-2-deoxyribose, β-D-ribose
(C) β-D-ribose, β-D-2-deoxyribose.
(D) β-D-deoxyribose, β-D-2-deoxyribose.

Open solution

Q87. JEE Main 2022 (Online) 28th June Evening Shift

When sugar 'X' is boiled with dilute H_2SO_4 in alcoholic solution, two isomers 'A' and 'B' are formed. 'A' on oxidation with HNO_3 yields saccharic acid where as 'B' is laevorotatory. The compound 'X' is :

(A) Maltose
(B) Sucrose
(C) Lactose
(D) Starch

Open solution

Q88. JEE Main 2022 (Online) 28th June Morning Shift

Stability of α-Helix structure of proteins depends upon

(A) dipolar interaction
(B) H-bonding interaction
(C) van der Waals forces
(D) π-stacking interaction

Open solution

Q89. JEE Main 2022 (Online) 27th June Evening Shift

Given below are two statements.

Statement I : Maltose has two α-D-glucose units linked at C_1 and C_4 and is a reducing sugar.

Statement II : Maltose has two monosaccharides : α-D-glucose and β-D-glucose linked at C_1 and C_6 and it is a non-reducing sugar.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true.
(B) Both Statement I and Statement II are false.
(C) Statement I is true but Statement II is false.
(D) Statement I is false but Statement II is true.

Open solution

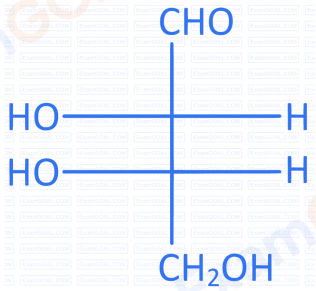
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Q90. JEE Main 2022 (Online) 27th June Morning Shift

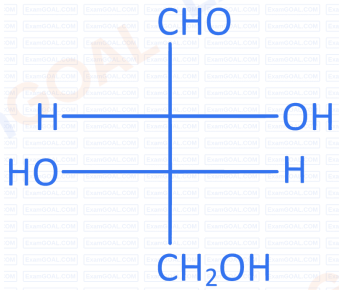
Open solution

L-isomer of a compound 'A' ($C_4H_8O_4$) gives a positive test with $[Ag(NH_3)_2]^+$. Treatment of 'A' with acetic anhydride yields triacetate derivative. Compound 'A' produces an optically active compound (B) and an optically inactive compound (C) on treatment with bromine water and HNO_3 respectively. Compound (A) is :

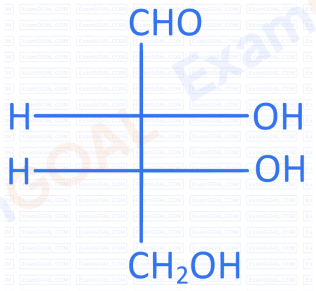
(A)



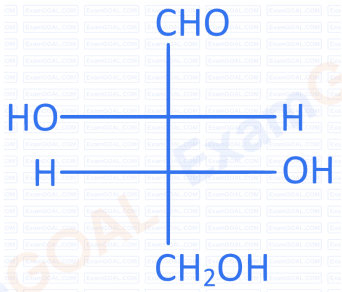
(B)



(C)



(D)



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Q91. JEE Main 2022 (Online) 26th June Evening Shift

Match List-I with List-II

	List - I Enzyme		List - II Conversion of
A.	Invertase	I.	Starch into maltose
B.	Zymase	II.	Maltose into glucose
C.	Diastase	III.	Glucose into ethanol
D.	Maltase	IV.	Cane sugar into glucose

Choose the most appropriate answer from the options given below :

(A) A-III, B-IV, C-II, D-I
(B) A-III, B-II, C-I, D-IV
(C) A-IV, B-III, C-I, D-II
(D) A-IV, B-II, C-III, D-I

Open solution

Q92. JEE Main 2022 (Online) 26th June Evening Shift

Which one of the following is a water soluble vitamin, that is not excreted easily?

(A) Vitamin B_2
(B) Vitamin B_1
(C) Vitamin B_6
(D) Vitamin B_{12}

Open solution

Q93. JEE Main 2022 (Online) 24th June Morning Shift

A polysaccharide 'X' on boiling with dil H_2SO_4 at 393 K under 2-3 atm pressure yields 'Y'. 'Y' on treatment with bromine water gives gluconic acid. 'X' contains β-glycosidic linkages only. Compound 'X' is :

(A) starch
(B) cellulose
(C) amylose
(D) amylopectin

Open solution

Q94. JEE Main 2021 (Online) 31st August Evening Shift

Which of the following is not an example of fibrous protein?

(A) Keratin
(B) Albumin
(C) Collagen
(D) Myosin

Open solution

Q95. JEE Main 2021 (Online) 31st August Morning Shift

Which one of the following compounds contains β-C_1-C_4 glycosidic linkage?

(A) Lactose
(B) Sucrose
(C) Maltose
(D) Amylose

Open solution

Q96. JEE Main 2021 (Online) 27th August Evening Shift

Hydrolysis of sucrose gives :

(A) α-D-(−)-Glucose and β-D-(−)-Fructose
(B) α-D-(+)-Glucose and α-D-(−)-Fructose
(C) α-D-(−)-Glucose and α-D-(+)-Fructose
(D) α-D-(+)-Glucose and β-D-(−)-Fructose

Open solution

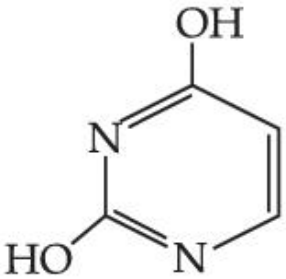
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Q97. JEE Main 2021 (Online) 27th August Morning Shift

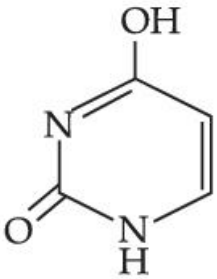
Out of following isomeric forms of uracil, which one is present in RNA?

Open solution

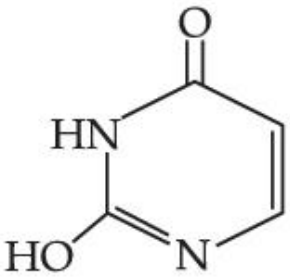
(A)



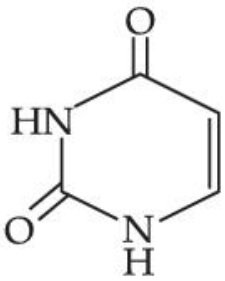
(B)



(C)



(D)



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Q98. JEE Main 2021 (Online) 26th August Evening Shift

Open solution

Given below are two statements : one is labelled as Assertion (A) and other is labelled as Reason (R).

Assertion (A) : Sucrose is a disaccharide and a non-reducing sugar.

Reason (R) : Sucrose involves glycosidic linkage between C_1 of β-glucose and C_2 of α-fructose.

Choose the most appropriate answer from the options given below :

(A) Both (A) and (R) are true but (R) is not the true explanation of (A).

(B) (A) is false but (R) is true.

(C) (A) is true but (R) is false.

(D) Both (A) and (R) are true and (R) is the true explanation of (A)

Q99. JEE Main 2021 (Online) 27th July Evening Shift

Open solution

Compound A gives D-Galactose and D-Glucose on hydrolysis. The compound A is :

(A) Amylose **(B)** Sucrose **(C)** Maltose **(D)** Lactose

Q100. JEE Main 2021 (Online) 27th July Morning Shift

Open solution

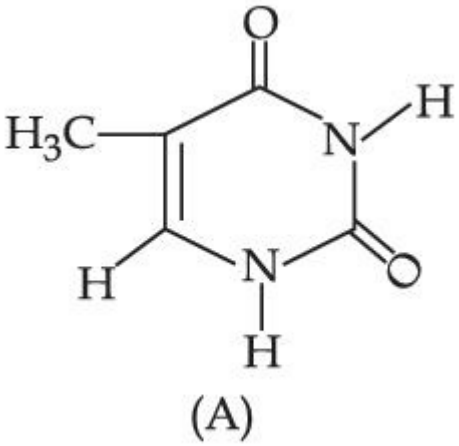
Which one among the following chemical tests is used to distinguish monosaccharide from disaccharide?

(A) Seliwanoff's test **(B)** Iodine test

(C) Barfoed test **(D)** Tollens' test

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Q101. JEE Main 2021 (Online) 27th July Morning Shift



(A)

Open solution

The compound 'A' is a complementary base of _____ in DNA strands.

(A) Uracil **(B)** Guanine **(C)** Adenine **(D)** Cytosine

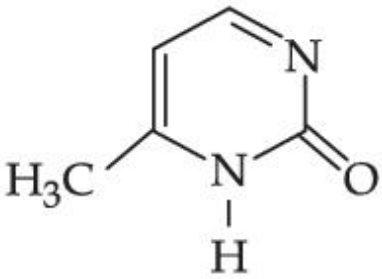
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Q102. JEE Main 2021 (Online) 25th July Evening Shift

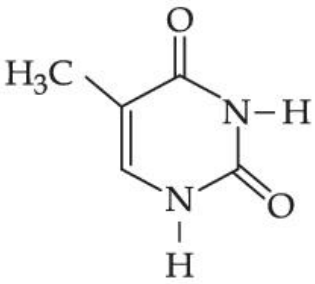
Open solution

Which one of the following is correct structure for cytosine?

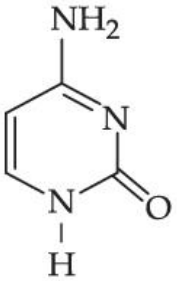
(A)



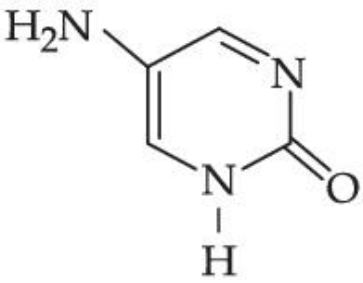
(B)



(C)



(D)



Q103. JEE Main 2021 (Online) 25th July Morning Shift

Open solution

The water soluble protein is :

(A) Fibrin (B) Albumin (C) Myosin (D) Collagen

Q104. JEE Main 2021 (Online) 22th July Evening Shift

Open solution

Thiamine and pyridoxine are also known respectively as :

(A) Vitamin B_2 and Vitamin E

(B) Vitamin E and Vitamin B_2

(C) Vitamin B_6 and Vitamin B_2

(D) Vitamin B_1 and Vitamin B_6

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Q105. JEE Main 2021 (Online) 20th July Evening Shift

Which one of the following statements is not true about enzymes?

(A) Enzymes are non-specific for a reaction and substrate.

(B) Almost all enzymes are proteins.

(C) Enzymes work as catalysts by lowering the activation energy of a biochemical reaction.

(D) The action of enzymes is temperature and pH specific.

Open solution

Q106. JEE Main 2021 (Online) 20th July Morning Shift

Identify the incorrect statement from the following

(A) Amylose is a branched chain polymer of glucose

(B) Starch is a polymer of α-D glucose

(C) β-Glycosidic linkage makes cellulose polymer

(D) Glycogen is called as animal starch

Open solution

Q107. JEE Main 2021 (Online) 18th March Evening Shift

Deficiency of vitamin K causes :

(A) Cheilosis

(B) Decrease in blood clotting time

(C) Increase in blood clotting time

(D) Increase in fragility of RBC's

Open solution

Q108. JEE Main 2021 (Online) 18th March Morning Shift

A non-reducing sugar "A" hydrolyses to give two reducing mono saccharides. Sugar A is

(A) Fructose

(B) Galactose

(C) Glucose

(D) Sucrose

Open solution

Q109. JEE Main 2021 (Online) 17th March Evening Shift

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{Enzyme\ A} \underset{Glu\cos e}{C_6H_{12}O_6} + \underset{Fructose}{C_6H_{12}O_6}$$

$$\underset{Glu\cos e}{C_6H_{12}O_6} \xrightarrow{Enzyme\ B} 2C_2H_5OH + 2CO_2$$

In the above reactions, the enzyme A and enzyme B respectively are :

(A) Zymase and Invertase

(B) Invertase and Amylase

(C) Amylase and Invertase

(D) Invertase and Zymase

Open solution

Q110. JEE Main 2021 (Online) 17th March Evening Shift

Fructose is an example of :

(A) Heptose

(B) Pyranose

(C) Ketohexose

(D) Aldohexose

Open solution

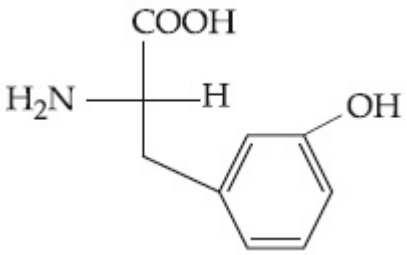
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Q111. JEE Main 2021 (Online) 17th March Morning Shift

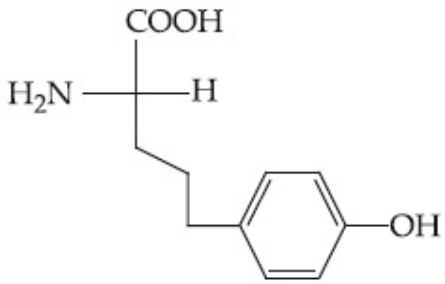
Which of the following is correct structure of tyrosine?

Open solution

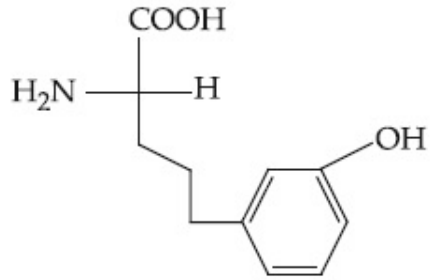
(A)



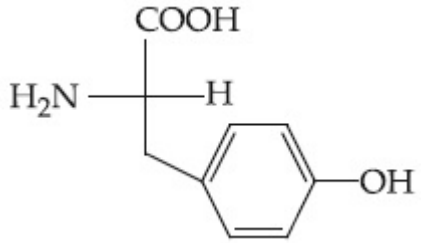
(B)



(C)



(D)



Q112. JEE Main 2021 (Online) 16th March Evening Shift

The secondary structure of protein is stabilised by:

(A) Hydrogen bonding
(B) glycosidic bond
(C) Peptide bond
(D) van der Waals forces

Open solution

Q113. JEE Main 2021 (Online) 16th March Morning Shift

Which among the following pairs of Vitamins is stored in our body relatively for longer duration?

(A) Thiamine and Vitamin A
(B) Ascorbic acid and Vitamin D
(C) Thiamine and Ascorbic acid
(D) Vitamin A and Vitamin D

Open solution

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Q114. JEE Main 2021 (Online) 26th February Evening Shift

Match List - I with List - II.

Open solution

	List - I		List - II
(a)	Sucrose	(i)	β-D-Galactose and β-D-Glucose
(b)	Lactose	(ii)	α-D-Glucose and β-D-Fructose
(c)	Maltose	(iii)	α-D-Glucose and α-D-Glucose

Choose the correct answer from the options given below :

(A) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (ii), (c) $\rightarrow$ (i)

(B) (a) $\rightarrow$ (i), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (ii)

(C) (a) $\rightarrow$ (ii), (b) $\rightarrow$ (i), (c) $\rightarrow$ (iii)

(D) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (i), (c) $\rightarrow$ (ii)

Q115. JEE Main 2021 (Online) 26th February Morning Shift

Which of the following vitamins is helpful in delaying the blood clotting?

Open solution

(A) Vitamin K **(B)** Vitamin C **(C)** Vitamin E **(D)** Vitamin B

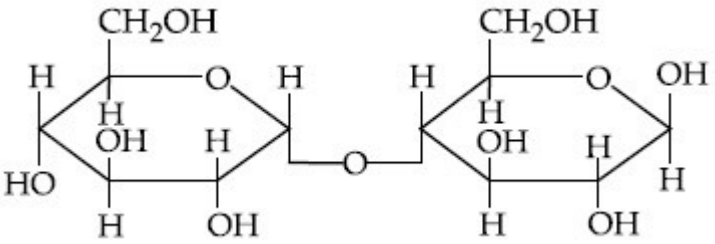
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Q116. JEE Main 2021 (Online) 25th February Evening Shift

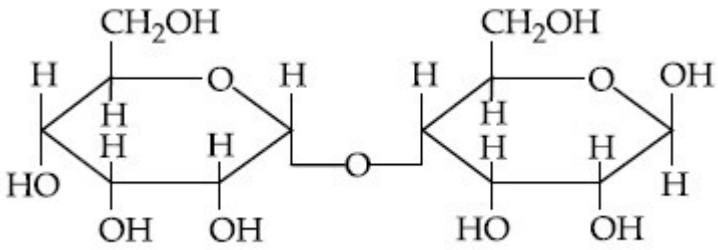
Open solution

Which of the following is correct structure of α-anomer of maltose?

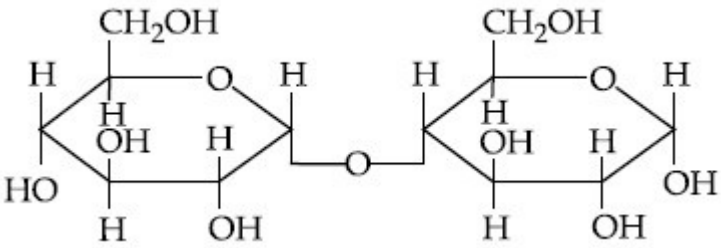
(A)



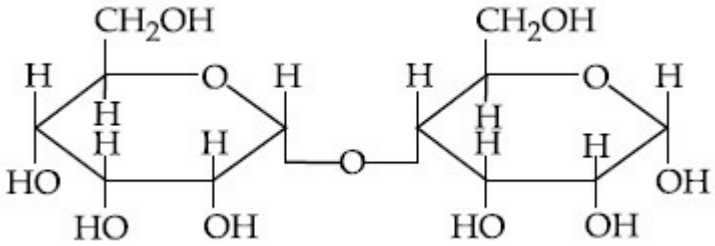
(B)



(C)



(D)



Q117. JEE Main 2021 (Online) 25th February Morning Shift

Open solution

Which of the glycosidic linkage between galactose and glucose is present in lactose?

(A) C-1 of galactose and C-6 of glucose

(B) C-1 of glucose and C-6 of galactose

(C) C-1 of galactose and C-4 of glucose

(D) C-1 of glucose and C-4 of galactose

Q118. JEE Main 2021 (Online) 24th February Morning Shift

Open solution

Out of the following, which type of interaction is responsible for the stabilisation of α-helix structure of proteins ?

(A) Covalent bonding

(B) van der Waals forces

(C) Ionic bonding

(D) Hydrogen bonding

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Q119. JEE Main 2020 (Online) 6th September Evening Slot

Which one of the following statements is not true?

(A) Lactose contains α-glycosidic linkage between C_1 of galactose and C_4 of glucose

(B) Lactose is a reducing sugar and it gives Fehling's test

(C) On acid hydrolysis, lactose gives one molecule of D(+)-glucose and one molecule of D(+)-galactose

(D) Lactose ($C_{12}H_{22}O_{11}$) is a disaccharide and it contains 8 hydroxyl groups

Open solution

Q120. JEE Main 2020 (Online) 5th September Morning Slot

Which of the following is not an essential amino acid :

(A) Valine (B) Leucine (C) Lysine (D) Tyrosine

Open solution

Q121. JEE Main 2020 (Online) 4th September Morning Slot

Which of the following will react with $CHCl_3$ + alc. KOH?

(A) Adenine and thymine (B) Thymine and proline

(C) Adenine and lysine (D) Adenine and proline

Open solution

Q122. JEE Main 2020 (Online) 4th September Morning Slot

What are the functional groups present in the structure of maltose?

(A) One acetal and one ketal (B) One ketal and one hemiketal

(C) Two acetals (D) One acetal and one hemiacetal

Open solution

Q123. JEE Main 2020 (Online) 2nd September Evening Slot

The correct observation in the following reactions is :

$$\text{Sucrose} \xrightarrow[\text{Cleavage (Hydrolysis)}]{\text{Glycosidic bond}} A + B \xrightarrow{\text{Seliwanoff's reagent}} ?$$

(A) Formation of blue colour (B) Formation of violet colour

(C) Formation of red colour (D) Gives no colour

Open solution

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Q124. JEE Main 2020 (Online) 2nd September Morning Slot

Open solution

Consider the following reactions:

(i) Glucose + ROH $\xrightarrow{\text{dry HCl}}$ Acetal $\xrightarrow[(CH_3CO)_2O]{\text{x eq.of}}$ acetyl derivative

(ii) Glucose $\xrightarrow{Ni/H_2}$ A $\xrightarrow[(CH_3CO)_2O]{\text{y eq.of}}$ acetyl derivative

(iii) Glucose $\xrightarrow[(CH_3CO)_2O]{\text{z eq.of}}$ acetyl derivative

'x', 'y' and 'z' in these reactions are respectively.

(A) 5, 6 & 5 (B) 4, 5 & 5 (C) 4, 6 & 5 (D) 5, 4 & 5

Q125. JEE Main 2020 (Online) 8th January Evening Slot

Open solution

Two monomers in maltose are :

(A) α-D-glucose and α-D-galactose (B) α-D-glucose and α-D-Fructose

(C) α-D-glucose and α-D-glucose (D) α-D-glucose and β-D-glucose

Q126. JEE Main 2020 (Online) 8th January Morning Slot

Open solution

Which of the following statement is not true for glucose?

(A) Glucose gives Schiff's test for aldehyde

(B) Glucose reacts with hydroxylamine to form oxime

(C) The pentaacetate of glucose does not react with hydroxylamine to give oxime

(D) Glucose exists in two crystalline forms α and β

Q127. JEE Main 2020 (Online) 7th January Evening Slot

Open solution

Which of the following statements is correct ?

(A) Gluconic acid can form cyclic (acetal/hemiacetal) structure

(B) Gluconic acid is a partial oxidation product of glucose

(C) Gluconic acid is obtained by oxidation of glucose with (HNO_3)

(D) Gluconic acid is a dicarboxylic acid.

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Q128. JEE Main 2020 (Online) 7th January Morning Slot

Open solution

Match the following :
(i) Riboflavin (a) Beriberi
(ii) Thiamine (b) Scurvy
(iii) Pyridoxine (c) Cheliosis
(iv) Ascorbic acid (d) Convulsions

(A) (i)-(c), (ii)-(d), (iii)-(a), (iv)-(b)
(B) (i)-(d), (ii)-(b), (iii)-(a), (iv)-(c)
(C) (i)-c, (ii)-(a), (iii)-(d), (iv)-(b)
(D) (i)-(a), (ii)-(d), (iii)-(c), (iv)-(b)

Q129. JEE Main 2019 (Online) 12th April Evening Slot

Open solution

Which of the given statements is **INCORRECT** about glycogen?

(A) It is a straight chain polymer similar to amylose
(B) It is present in animal cells
(C) It is present in some yeast and fungi
(D) Only α-linkages are present in the molecule

Q130. JEE Main 2019 (Online) 12th April Morning Slot

Open solution

Glucose and Galactose are having identical configuration in all the positions except position

(A) C−5 **(B)** C−3 **(C)** C−2 **(D)** C−4

Q131. JEE Main 2019 (Online) 12th April Morning Slot

Open solution

Which of the following statements is not true about RNA?

(A) It controls the synthesis of protein
(B) It is present in the nucleus of the cell
(C) It usually does not replicate
(D) It has always double stranded α-helix structure

Q132. JEE Main 2019 (Online) 10th April Evening Slot

Open solution

Number of stereo centers present in linear and cyclic structures of glucose are respectively :

(A) 4 & 4 **(B)** 4 & 5 **(C)** 5 & 4 **(D)** 5 & 5

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Q133. JEE Main 2019 (Online) 10th April Morning Slot

Amylopectin is compound of :

(A) α-D-glucose, $C_1 - C_4$ and $C_1 - C_6$ linkages

(B) β-D-glucose, $C_1 - C_4$ and $C_2 - C_6$ linkages

(C) β-D-glucose, $C_1 - C_4$ and $C_1 - C_6$ linkages

(D) α-D-glucose, $C_1 - C_4$ and $C_2 - C_6$ linkages

Open solution

Q134. JEE Main 2019 (Online) 9th April Evening Slot

The peptide that gives positive ceric ammonium nitrate and carbylamine tests is :

(A) Ser-Lys (B) Asp-Gln (C) Gln-Asp (D) Lys-Asp

Open solution

Q135. JEE Main 2019 (Online) 9th April Morning Slot

Which of the following statements is not true about sucrose?

(A) The glycosidic linkage is present between C_1 of α-glucose and C_1 of β-fructose

(B) It is also named as invert sugar

(C) It is a non reducing sugar

(D) On hydrolysis, it produces glucose and fructose

Open solution

Q136. JEE Main 2019 (Online) 8th April Evening Slot

Fructose and glucose can be distinguished by :

(A) Barfoed's test (B) Fehling's test

(C) Seliwanoff's test (D) Benedict's test

Open solution

Q137. JEE Main 2019 (Online) 8th April Morning Slot

Maltose on treatment with dilute HCl gives :

(A) D-Galactose (B) D-Glucose and D-Fructose

(C) D-Glucose (D) D-Fructose

Open solution

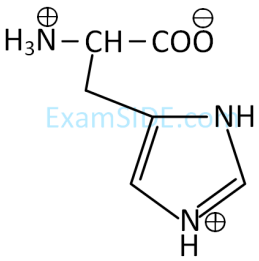
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Q138. JEE Main 2019 (Online) 12th January Evening Slot

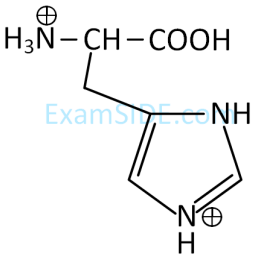
Open solution

The correct structure of histidine in a strongly acidic solution (pH = 2) is -

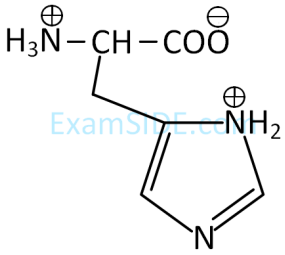
(A)



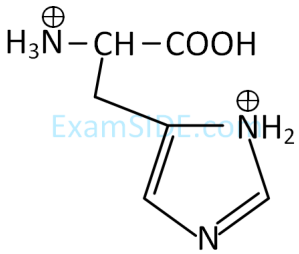
(B)



(C)



(D)



Q139. JEE Main 2019 (Online) 12th January Evening Slot

Open solution

The correct statement (s) among I to III with respect to potassium ions that are abundant within the cell fluid is /are

I. They activate many enzymes

II. They participate in the oxidation of glucose to produce ATP

III. Along with sodium ions they are responsible for the transmission of nerve signals

(A) III only

(B) I and II only

(C) I and III only

(D) I, II and III only

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Q140. JEE Main 2019 (Online) 12th January Morning Slot

Among the following compounds most basic amino acid is -

(A) Lysine **(B)** Asparagine **(C)** Histidine **(D)** Serine

Open solution

Q141. JEE Main 2019 (Online) 11th January Evening Slot

The correct match between Item I and Item II is :

Item I		Item II	
(A)	Ester test	(P)	Tyr
(B)	Carbylamine test	(Q)	Asp
(C)	Phthalein dye test	(R)	Ser
		(S)	Lys

(A) (A)→(Q); (B)→(S); (C)→(P)

(B) (A)→(R); (B)→(S); (C)→(Q)

(C) (A)→(R); (B)→(Q); (C)→(P)

(D) (A)→(Q); (B)→(S); (C)→(R)

Open solution

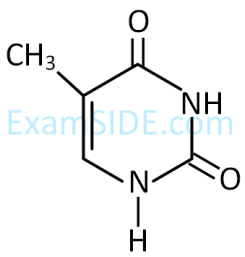
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Q142. JEE Main 2019 (Online) 11th January Morning Slot

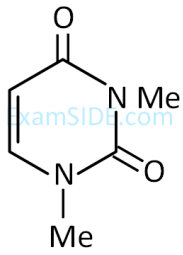
Open solution

Among the following compounds, which one is found in RNA ?

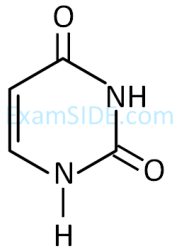
(A)



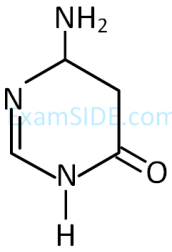
(B)



(C)



(D)



Q143. JEE Main 2019 (Online) 10th January Evening Slot

Open solution

Which of the following tests cannot be used for identifying amino acids?

(A) Ninhydrin test
(B) Barfoed test
(C) Xanthoproteic test
(D) Biuret test

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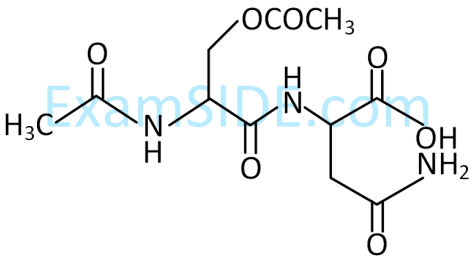
Q144. JEE Main 2019 (Online) 10th January Morning Slot

The correct structure of product 'P' in the following reactions is :

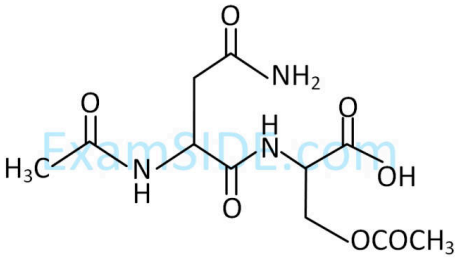
Open solution

$$\text{Asn} - \text{Ser} + \underset{\text{(excess)}}{(CH_3CO)_2O} \xrightarrow{NEt_3} P$$

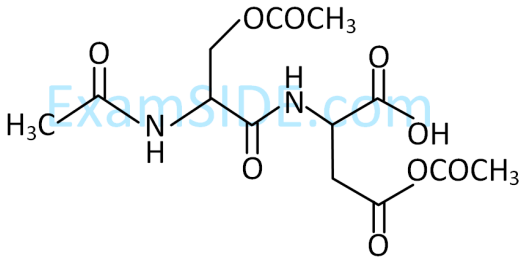
(A)



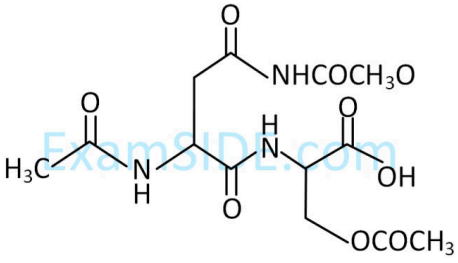
(B)



(C)



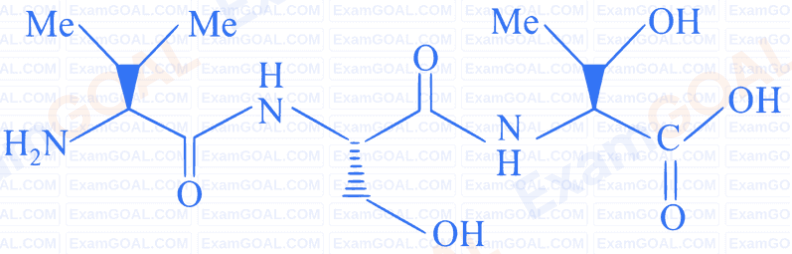
(D)



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Q145. JEE Main 2019 (Online) 9th January Evening Slot

The correct sequence of amino acids present in the tripeptide given below is :



Open solution

(A) Val - Ser - Thr (B) Thr - Ser - Val (C) Leu - Ser - Thr (D) Thr - Ser - Leu

Q146. JEE Main 2019 (Online) 9th January Morning Slot

The increasing order of pka of the following amino acids in aqueous solution is :

Gly, Asp, Lys, Arg

Open solution

(A) Asp < Gly < Arg < Lys (B) Gly < Asp < Arg < Lys

(C) Asp < Gly < Lys < Arg (D) Arg < Lys < Gly < Asp

Q147. JEE Main 2018 (Online) 16th April Morning Slot

Among the following, the **incorrect** statement is :

(A) Maltose and lactose has 1, 4 - glycosidic linkage.

(B) Sucrose and amylose has 1, 2 - glycosidic linkage

(C) Cellulose and amylose has 1, 4 - glycosidic linkage.

(D) Lactose contains β-D-galactose and β-D-glucose.

Open solution

Q148. JEE Main 2018 (Offline)

Glucose on prolonged heating with HI gives

(A) 6-iodohexanal (B) n-Hexane (C) 1-Hexene (D) Hexanoic acid

Open solution

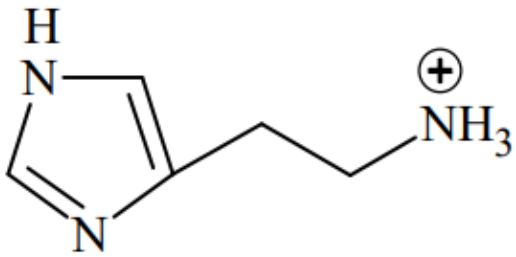
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Q149. JEE Main 2018 (Offline)

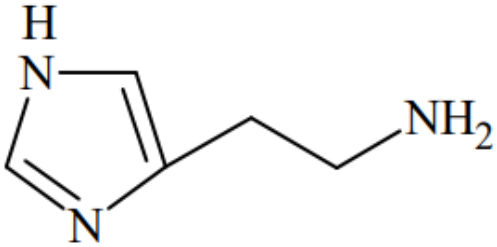
Open solution

The predominant form of histamine present in human blood is (pK_a, Histidine = 6.0)

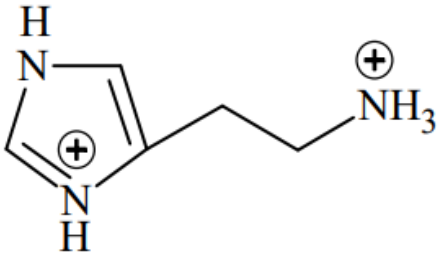
(A)



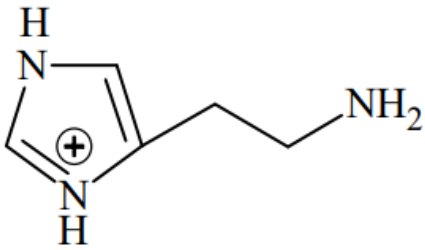
(B)



(C)



(D)



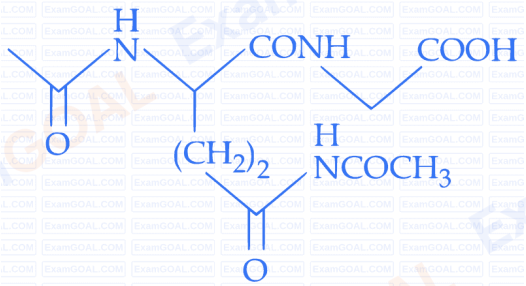
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Q150. JEE Main 2018 (Online) 15th April Evening Slot

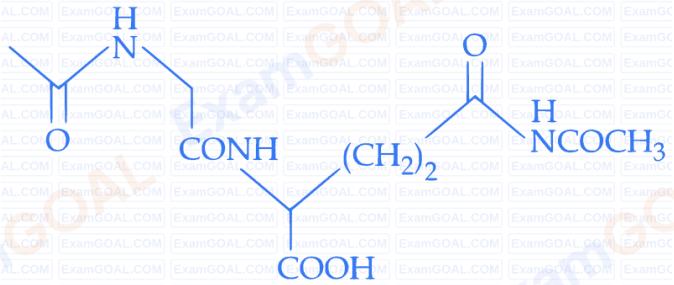
The dipeptide, Gln-Gly, on treatment with CH_3COCl followed by aqueous work up gives :

Open solution

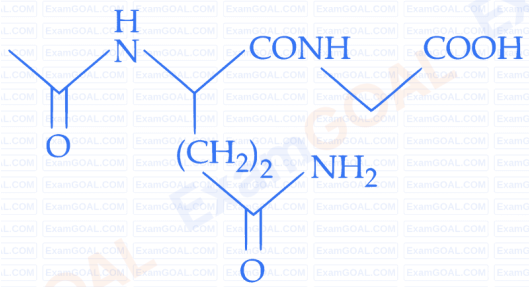
(A)



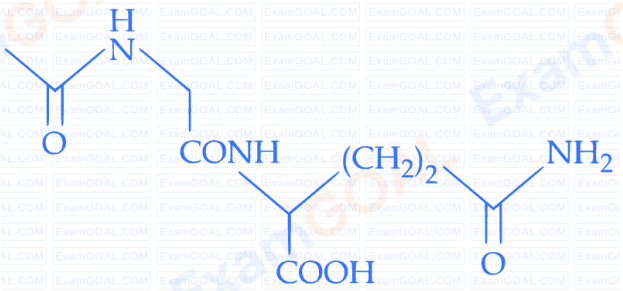
(B)



(C)



(D)



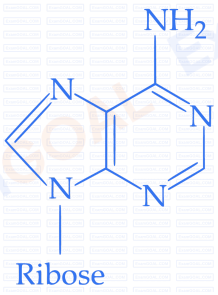
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Q151. JEE Main 2018 (Online) 15th April Morning Slot

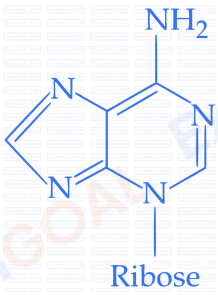
Open solution

Which of the following is the correct structure of Adenosine?

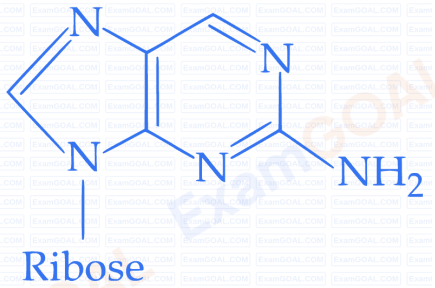
(A)



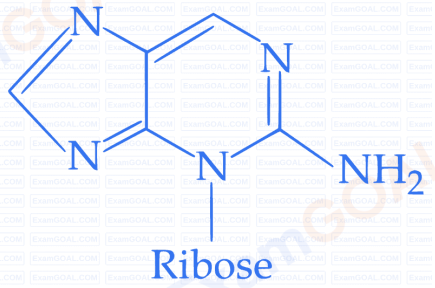
(B)



(C)



(D)



Q152. JEE Main 2017 (Online) 9th April Morning Slot

Open solution

The **incorrect** statement among the following is :

(A) α-D-glucose and β-D-glucose are anomers.

(B) α-D-glucose and β-D-glucose are enantiomers.

(C) Cellulose is a straight chain polysaccharide made up of only β-D-glucose units.

(D) The penta acetate of glucose does not react with hydroxyl amine

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Q153. JEE Main 2017 (Online) 8th April Morning Slot

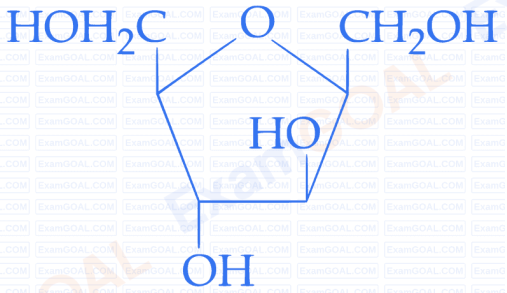
Among the following, the essential amino acid is :

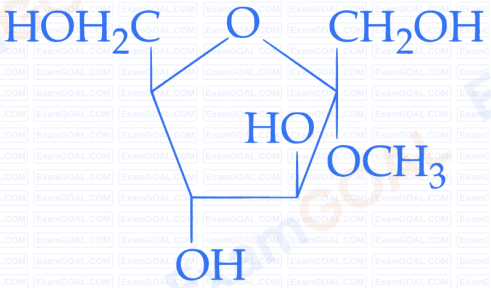
(A) Alanine (B) Valine (C) Aspartic acid (D) Serine

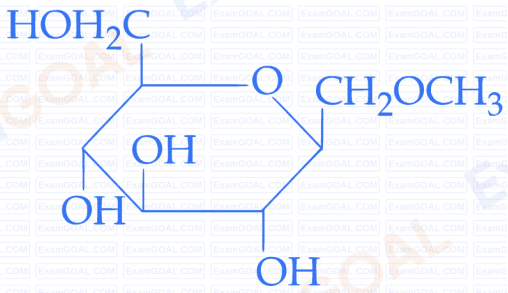
Open solution

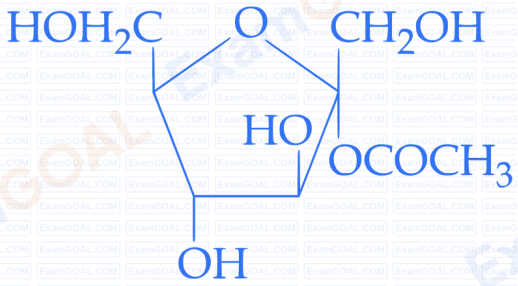
Q154. JEE Main 2017 (Offline)

Which of the following compounds will behave as a reducing sugar in an aqueous KOH solution?

(A) 

(B) 

(C) 

(D) 

Open solution

Q155. JEE Main 2016 (Online) 10th April Morning Slot

Observation of "Rhumann's purple" is a confirmatory test for the presence of :

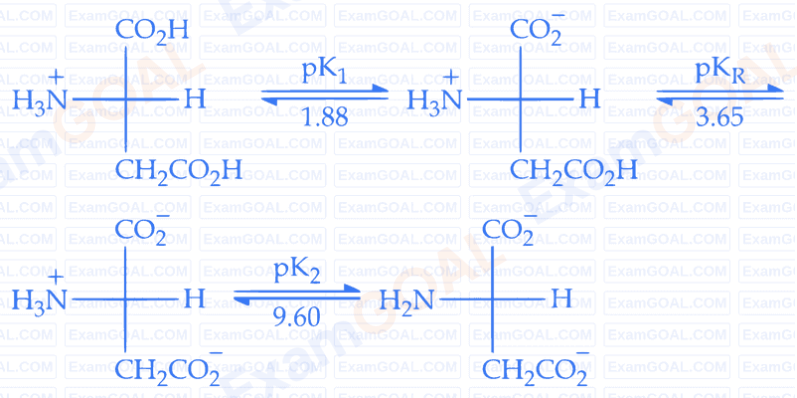
(A) Reducing sugar (B) Cupric ion (C) Protein (D) Starch

Open solution

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Q156. JEE Main 2016 (Online) 9th April Morning Slot

Consider the following sequence for aspartic acid :



The *pI* (isoelectric point) of aspartic acid is :

(A) 1.88 (B) 3.65 (C) 5.74 (D) 2.77

Open solution

Q157. JEE Main 2016 (Offline)

Thiol group is present in :

(A) Cytosine (B) Cystine (C) Cysteine (D) Methionine

Open solution

Q158. JEE Main 2015 (Online) 11th April Morning Slot

Accumulation of which of the following molecules in the muscles occurs as a result of vigorous exercise ?

(A) Glucose (B) Glycogen (C) L-lactic acid (D) Pyruvic acid

Open solution

Q159. JEE Main 2015 (Online) 10th April Morning Slot

Complete hydrolysis of starch gives :

(A) glucose and fructose in equimolar amounts

(B) galactose and fructose in equimolar amounts

(C) glucose only

(D) glucose and galactose in equimolar amounts

Open solution

Q160. JEE Main 2015 (Offline)

Which of the vitamins given below is water soluble?

(A) Vitamin D (B) Vitamin E (C) Vitamin K (D) Vitamin C

Open solution

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Q161. JEE Main 2014 (Online) 19th April Morning Slot

The reason for double helical structure of DNA is the operation of :

(A) Electrostatic attractions
(B) van der Waals forces
(C) Dipole - Dipole interactions
(D) Hydrogen bonding

Open solution

Q162. JEE Main 2014 (Online) 12th April Morning Slot

Which of the following will **not** show mutarotation ?

(A) Maltose (B) Lactose (C) glucose (D) Sucrose

Open solution

Q163. JEE Main 2014 (Offline)

Which one of the following bases is **not** present in DNA?

(A) Cytosine (B) Thymine (C) Quinoline (D) Adenine

Open solution

Q164. JEE Main 2013 (Online) 25th April Morning Slot

Which of the following statement is not correct?

(A) Amylopectin is a branched polymer of α-glucose.
(B) Cellulose is a linear polymer of β-glucose.
(C) Glycogen is the food reserve of plants.
(D) All proteins are polymers of α-amino acids.

Open solution

Q165. JEE Main 2013 (Online) 25th April Morning Slot

Among the following vitamins the one whose deficiency causes rickets (bone deficiency) is :

(A) Vitamin A (B) Vitamin B (C) Vitamin D (D) Vitamin C

Open solution

Q166. JEE Main 2013 (Online) 23th April Morning Slot

Glycosidic linkage is actually an :

(A) Carbonyl bond (B) Ether bond (C) Ester bond (D) Amide bond

Open solution

Q167. JEE Main 2013 (Online) 23th April Morning Slot

Natural glucose is termed D-glucose because :

(A) -OH on the second carbon is on the right side in Fischer projection
(B) -OH on the sixth carbon is on the right side in Fischer projection.
(C) -OH on the fifth carbon is on the right side in Fischer projection.
(D) It is dextrorotatory.

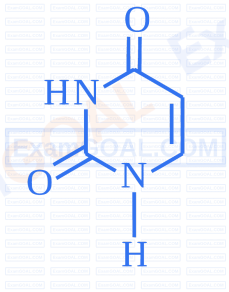
Open solution

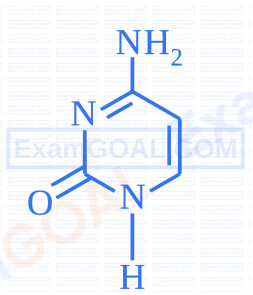
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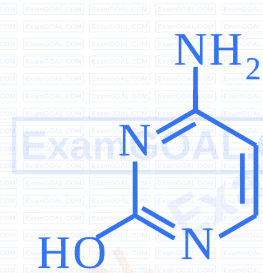
Q168. JEE Main 2013 (Online) 22th April Morning Slot

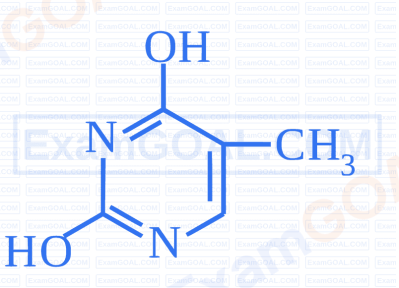
Open solution

Which of the following structures represents thymine?

(A) 

(B) 

(C) 

(D) 

Q169. JEE Main 2013 (Online) 9th April Morning Slot

Open solution

Which of the following enzyme converts starch into maltose?

(A) Diastase (B) Maltase (C) Zymase (D) Invertase

Q170. JEE Main 2013 (Offline)

Open solution

Synthesis of each molecule of glucose in photosynthesis involves:

(A) 10 molecules of ATP (B) 8 molecules of ATP

(C) 6 molecules of ATP (D) 18 molecules of ATP

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Q171. AIEEE 2012 (Online) 26th May

Which of the following statements is correct?

(A) RNA controls the synthesis of proteins.

(B) The sugar present in DNA is D-(−)-ribose.

(C) RNA has double stranded α-helix structure.

(D) DNA mainly occurs in the cytoplasm of the cell.

Open solution

Q172. AIEEE 2012 (Online) 19th May

Which of the following is a non-reducing sugar?

(A) Lactose (B) Fructose (C) Sucrose (D) Maltose

Open solution

Q173. AIEEE 2012 (Online) 12th May

Amylopectin is a polymer of

(A) α-D-glucose (B) amino acid (C) β-D-glucose (D) amylase.

Open solution

Q174. AIEEE 2012 (Online) 7th May

All of the following statements apply to proteins except

(A) Proteins generally have no definite melting point

(B) Proteins contain the grouping $-\mathrm{CONH}-$

(C) Proteins have high molecular weight

(D) Proteins can only contain the elements C, H, O and N.

Open solution

Q175. AIEEE 2012

Which of the following compounds can be detected by Molisch's test ?

(A) Nitro compounds (B) Sugars (C) Amines (D) Primary alcohols

Open solution

Q176. AIEEE 2012

Which one of the following statements is correct ?

(A) All amino acids except lysine are optically active

(B) All amino acids are optically active

(C) All amino acids except glycine are optically active

(D) All amino acids except glutamic acid are optically active

Open solution

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Q177. AIEEE 2011

The presence or absence of hydroxyl group on which carbon atom of sugar differentiates RNA and DNA ?

(A) 2nd (B) 3rd (C) 4th (D) 1st

Open solution

Q178. AIEEE 2010

Biuret test is **not** given by

(A) carbohydrates (B) polypeptides (C) urea (D) proteins

Open solution

Q179. AIEEE 2009

The two functional groups present in a typical carbohydrate are

(A) -OH and -COOH (B) -CHO and -COOH (C) > C = O and - OH (D) - OH and -CHO

Open solution

Q180. AIEEE 2008

α-D-(+)-glucose and β-D-(+)-glucose are

(A) conformers (B) epimers (C) anomers (D) enantiomers

Open solution

Q181. AIEEE 2007

The secondary structure of a protein refers to

(A) α-helical backbone

(B) hydrophobic interactions

(C) sequence of α-amino acids

(D) fixed configuration of the polypeptide backbone

Open solution

Q182. AIEEE 2006

The term anomers of glucose refers to

(A) isomers of glucose that differ in configurations at carbons one and four (C-1 and C-4)

(B) a mixture of (D)-glucose and (L)-glucose

(C) enantiomers of glucose

(D) isomers of glucose that differ in configuration at carbon one (C-1)

Open solution

Q183. AIEEE 2006

The pyrimidine bases present in DNA are

(A) cytosine and adenine (B) cytosine and guanine

(C) cytosine and thymine (D) cytosine and uracil

Open solution

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Q184. AIEEE 2005

In both DNA and RNA, heterocyclic base and phosphate ester linkages are at-

(A) C_5' and C_2' respectively of the sugar molecule

(B) C_2' and C_5' respectively of the sugar molecule

(C) C_1' and C_5' respectively of the sugar molecule

(D) C_5' and C_1' respectively of the sugar molecule

Open solution

Q185. AIEEE 2004

Identify the correct statements regarding enzymes

(A) Enzymes are specific biological catalysts that can normally function at very high temperature (T ~ 1000 K)

(B) Enzymes are specific biological catalysts that the posses well – defined active sites

(C) Enzymes are specific biological catalysts that can not be poisoned

(D) Enzymes are normally heterogeneous catalysts that are very specific in their action

Open solution

Q186. AIEEE 2004

Which base is present in RNA but not in DNA?

(A) Uracil (B) Thymine (C) Guanine (D) Cytosine

Open solution

Q187. AIEEE 2004

Insulin production and its action in human body are responsible for the level of diabetes. This compound belongs to which of the following categories?

(A) A co- enzyme (B) An antibiotic (C) An enzyme (D) A hormone

Open solution

Q188. AIEEE 2003

Complete hydrolysis of cellulose gives

(A) D-ribose (B) D-glucose (C) L-glucose (D) D-fructose

Open solution

Q189. AIEEE 2003

The reason for double helical structure of DNA is operation of

(A) dipole-dipole interaction (B) hydrogen bonding

(C) electrostatic attractions (D) van der Waals' forces

Open solution

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Q190. AIEEE 2002

The functional group, which is found in amino acid is

(A) -COOH group (B) $-NH_2$ group (C) $-CH_3$ group (D) both (a) and (b)

Open solution

Q191. AIEEE 2002

RNA is different from DNA because RNA contains

(A) ribose sugar and thymine (B) ribose sugar and uracil

(C) deoxyribose sugar and thymine (D) deoxyribose sugar and uracil

Open solution

Numerical 23 questions

Q192. JEE Main 2025 (Online) 4th April Morning Shift

The total number of hydrogen bonds of a DNA-double Helix strand whose one strand has the following sequence of bases is _____.

$5' - G - G - C - A - A - A - T - C - G - G - C - T - A - 3'$

Open solution

Q193. JEE Main 2024 (Online) 9th April Morning Shift

Total number of essential amino acid among the given list of amino acids is _____.

Arginine, Phenylalanine, Aspartic acid, Cysteine, Histidine, Valine, Proline

Open solution

Q194. JEE Main 2024 (Online) 8th April Evening Shift

The total number of carbon atoms present in tyrosine, an amino acid, is _____.

Open solution

Q195. JEE Main 2024 (Online) 1st February Evening Shift

The number of tripeptides formed by three different amino acids using each amino acid once is _____.

Open solution

Q196. JEE Main 2024 (Online) 31st January Evening Shift

From the vitamins $A, B_1, B_6, B_{12}, C, D, E$ and K, the number of vitamins that can be stored in our body is _____.

Open solution

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Q197. JEE Main 2024 (Online) 30th January Evening Shift

The total number of correct statements, regarding the nucleic acids is _____.

A. RNA is regarded as the reserve of genetic information

B. DNA molecule self-duplicates during cell division

C. DNA synthesizes proteins in the cell

D. The message for the synthesis of particular proteins is present in DNA

E. Identical DNA strands are transferred to daughter cells.

Open solution

Q198. JEE Main 2024 (Online) 29th January Morning Shift

Number of compounds among the following which contain sulphur as heteroatom is _____.

Furan, Thiophene, Pyridine, Pyrrole, Cysteine, Tyrosine

Open solution

Q199. JEE Main 2023 (Online) 12th April Morning Shift

In an oligopeptide named Alanylglycylphenyl alanyl isoleucine, the number of sp^2 hybridised carbons is _____.

Open solution

Q200. JEE Main 2023 (Online) 11th April Evening Shift

Number of compounds from the following which will not produce orange red precipitate with Benedict solution is _____.

Glucose, maltose, sucrose, ribose, 2-deoxyribose, amylose, lactose

Open solution

Q201. JEE Main 2023 (Online) 30th January Evening Shift

A short peptide on complete hydrolysis produces 3 moles of glycine (G), two moles of leucine (L) and two moles of valine (V) per mole of peptide. The number of peptide linkages in it are _____.

Open solution

Q202. JEE Main 2023 (Online) 24th January Evening Shift

Total number of tripeptides possible by mixing of valine and proline is _____

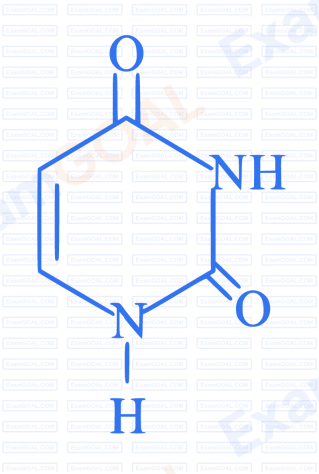
Open solution

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Q203. JEE Main 2023 (Online) 24th January Morning Shift

Uracil is a base present in RNA with the following structure. % of N in uracil is _____

Open solution



Given:

Molar mass N = 14 g mol^{-1}

O = 16 g mol^{-1}

C = 12 g mol^{-1}

H = 1 g mol^{-1}

Q204. JEE Main 2022 (Online) 29th July Morning Shift

In a linear tetrapeptide (Constituted with different amino acids), (number of amino acids) − (number of peptide bonds) is _____.

Open solution

Q205. JEE Main 2022 (Online) 29th June Morning Shift

$C_6H_{12}O_6 \xrightarrow{\text{Zymase}} A \xrightarrow[\Delta]{\text{NaOI}} B + CHI_3$

The number of carbon atoms present in the product B is _____.

Open solution

Q206. JEE Main 2022 (Online) 26th June Morning Shift

The number of oxygens present in a nucleotide formed from a base, that is present only in RNA is _____.

Open solution

Q207. JEE Main 2022 (Online) 25th June Evening Shift

How many of the given compounds will give a positive Biuret test _____ ?

Glycine, Glycylalanine, Tripeptide, Biuret

Open solution

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Q208. JEE Main 2022 (Online) 24th June Evening Shift

In alanylglycyl leucyl alanyl valine, the number of peptide linkages is _____.

Open solution

Q209. JEE Main 2021 (Online) 1st September Evening Shift

A peptide synthesized by the reactions of one molecule each of Glycine, Leucine, Aspartic acid and Histidine will have _____ peptide linkages.

Open solution

Q210. JEE Main 2021 (Online) 26th August Morning Shift

The total number of negative charge in the tetrapeptide, Gly-Glu-Asp-Tyr, at pH 12.5 will be _____. (Integer answer)

Open solution

Q211. JEE Main 2020 (Online) 5th September Evening Slot

The number of chiral carbons present in sucrose is ____.

Open solution

Q212. JEE Main 2020 (Online) 5th September Morning Slot

The number of chiral carbon(s) present in peptide, Ile-Arg-Pro, is ____.

Open solution

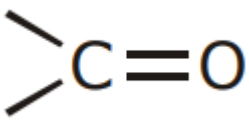
Q213. JEE Main 2020 (Online) 4th September Evening Slot

The number of chiral centres present in threonine is _____.

Open solution

Q214. JEE Main 2020 (Online) 3rd September Evening Slot

The number of



groups present in a tripeptide Asp–Glu–Lys is ___.

Open solution

Out of syllabus 2 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q215. JEE Main 2023 (Online) 6th April Morning Shift · MCQ (Single Correct Answer)

Open solution

Match List I with List II

	LIST I Vitamin		LIST II Deficiency disease
A.	Vitamin A	I.	Beri-Beri
B.	Thiamine	II.	Cheilosis
C.	Ascorbic acid	III.	Xeropthalmia
D.	Riboflavin	IV.	Scurvy

Choose the correct answer from the options given below:

(A) A-III, B-I, C-IV, D-II

(B) A-IV, B-I, C-III, D-II

(C) A-IV, B-II, C-III, D-I

(D) A-III, B-II, C-IV, D-I

Q216. JEE Main 2022 (Online) 25th June Evening Shift · MCQ (Single Correct Answer)

Open solution

Match List I with List II

	List - I		List -II
A.	Zymase	I.	Stomach
B.	Diastase	II.	Yeast
C.	Urease	III.	Malt
D.	Pepsin	IV.	Soyabean

Choose the correct answer from the options given below :

(A) A-II, B-III, C-I, D-IV

(B) A-II, B-III, C-IV, D-I

(C) A-III, B-II, C-IV, D-I

(D) A-III, B-II, C-I, D-IV

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **C**	3. **C**	4. **B**	5. **C**
6. **A**	7. **D**	8. **C**	9. **D**	10. **A**
11. **C**	12. **C**	13. **A**	14. **D**	15. **D**
16. **B**	17. **D**	18. **C**	19. **D**	20. **B**
21. **C**	22. **B**	23. **B**	24. **D**	25. **C**
26. **C**	27. **B**	28. **B**	29. **B**	30. **A**
31. **D**	32. **A**	33. **C**	34. **C**	35. **B**
36. **C**	37. **D**	38. **C**	39. **A**	40. **B**
41. **B**	42. **A**	43. **A**	44. **D**	45. **A**
46. **C**	47. **B**	48. **D**	49. **B**	50. **A**
51. **B**	52. **D**	53. **C**	54. **B**	55. **A**
56. **C**	57. **A**	58. **B**	59. **D**	60. **A**
61. **D**	62. **A**	63. **C**	64. **B**	65. **B**
66. **D**	67. **D**	68. **B**	69. **C**	70. **C**
71. **D**	72. **D**	73. **A**	74. **C**	75. **D**
76. **A**	77. **D**	78. **A**	79. **A**	80. **C**
81. **A**	82. **B**	83. **A**	84. **A**	85. **C**
86. **B**	87. **B**	88. **B**	89. **C**	90. **A**
91. **C**	92. **D**	93. **B**	94. **B**	95. **A**
96. **D**	97. **D**	98. **C**	99. **D**	100. **C**
101. **C**	102. **C**	103. **B**	104. **D**	105. **A**
106. **A**	107. **C**	108. **D**	109. **D**	110. **C**
111. **D**	112. **A**	113. **D**	114. **C**	115. **A**
116. **C**	117. **C**	118. **D**	119. **A**	120. **D**
121. **C**	122. **D**	123. **C**	124. **C**	125. **C**
126. **A**	127. **B**	128. **C**	129. **A**	130. **D**
131. **D**	132. **B**	133. **A**	134. **A**	135. **A**
136. **C**	137. **C**	138. **B**	139. **D**	140. **A**
141. **A**	142. **C**	143. **B**	144. **B**	145. **A**
146. **C**	147. **B**	148. **B**	149. **A**	150. **C**

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151. **A**	152. **B**	153. **B**	154. **D**	155. **C**
156. **D**	157. **C**	158. **C**	159. **C**	160. **D**
161. **D**	162. **D**	163. **C**	164. **C**	165. **C**
166. **B**	167. **C**	168. **D**	169. **A**	170. **D**
171. **A**	172. **C**	173. **A**	174. **D**	175. **B**
176. **C**	177. **A**	178. **A**	179. **C**	180. **C**
181. **A**	182. **D**	183. **C**	184. **C**	185. **B**
186. **A**	187. **D**	188. **B**	189. **B**	190. **D**
191. **B**				

Numerical

192. **33**	193. **4**	194. **9**	195. **6**	196. **5**
197. **3**	198. **2**	199. **10**	200. **2**	201. **6**
202. **8**	203. **25**	204. **1**	205. **1**	206. **9**
207. **2**	208. **4**	209. **3**	210. **4**	211. **9**
212. **4**	213. **2**	214. **5**		

Out of syllabus

215. **A**	216. **B**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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